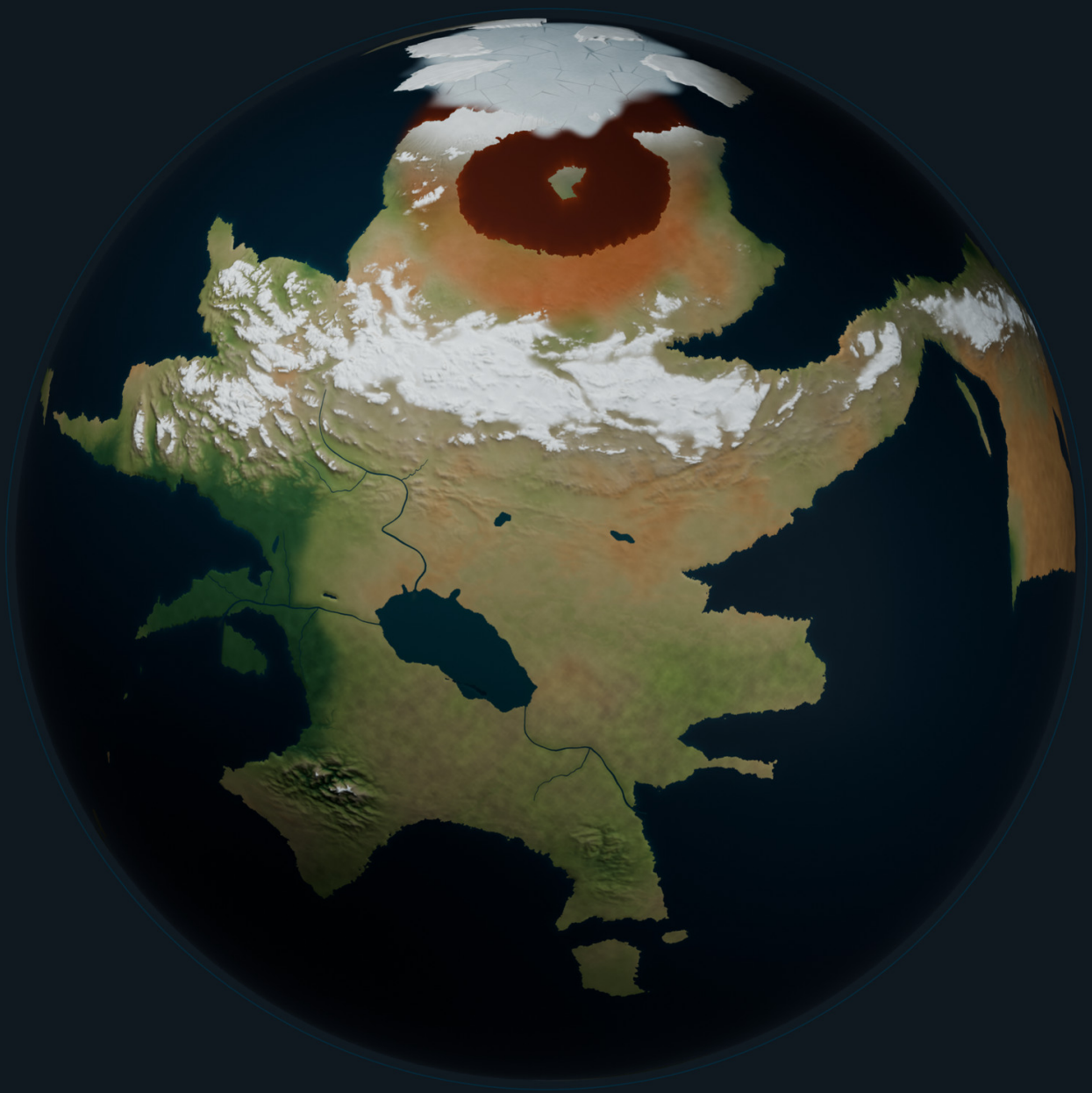
CATWORLD

THE WORLD MODEL

Creator & showrunner field guide



A planet to draw from. A geography to tell stories in.

Revision 8 / 31 October 2026

A separate southern continent and its eastern archipelago.
Northern autumn, southern spring, and an ancient crossing.

READING ROUTE

A guide to the model

Read the geography first, then use the reference plates. Established creator facts, draft-deck material and new cartographic proposals are distinguished throughout. Square-bracket source keys point to the reference notes.

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What is established, and what is proposed

Established geography

Terra has a 4,000-Earth-year orbit. The creator requested stronger axial tilt; this model uses a proposed 35 degrees. Panterra and the eastern continent share the giant mountain system. The separate southern mainland is closer to Panterra and has a sparse eastern island chain. No island arc links it to the eastern continent.

Current epoch

The north is in late autumn and its high country is already wintry. The south is in late spring approaching summer. The older southern-winter crossing and current southward bird movement remain the story framework.

Authoritative water order

The northern mountain river crosses the upper plateau, falls into the reservoir valley, and supplies the lake. A separate, slightly more southerly river drains west through Seattle country to the sea. The eastern megadam and its downstream city remain in place. This replaces the earlier Seattle-into-lake layout.

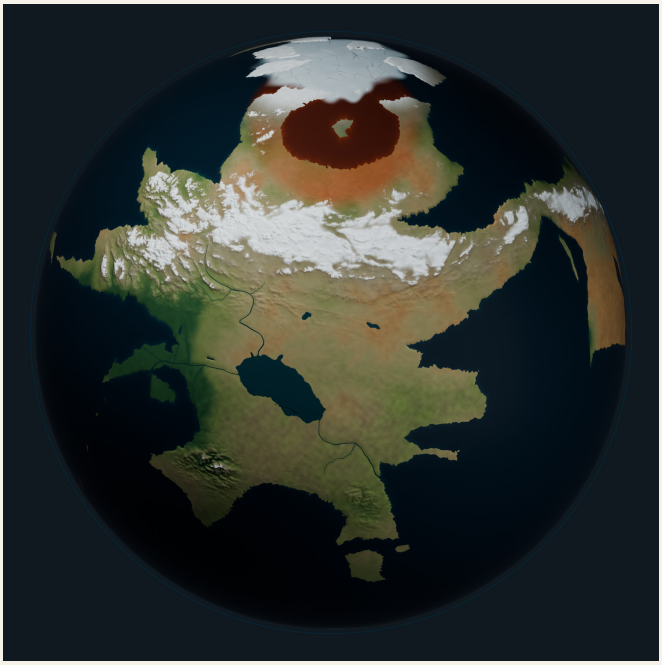
Regional environments

Most of northern Panterra is a broad mountain province. Steppe occupies the lower northern foothills. The west is extensive forest country around Seattle. Southern settled districts are fertile farmland and rolling hills.

Design proposals and limits

Radius 6,100 km, 35-degree tilt, 24-hour rotation, Ls 255 degrees, a 65 m former lowstand and an illustrative crossing 1,100 years ago remain proposals. Exact ice extent, heat source, water budget and engineering need further development. This is a consistent drawing-reference model, not a climate or engineering simulation.

The planet at a glance



PANTERRA / FRONT



FAR-SIDE HEMISPHERE

Three continental identities

Panterra is the main story landscape. The eastern dry core remains attached through the mountain belt, rotated eight degrees downward at its junction. A deep ocean separates it from the independent southern shield.

The southern silhouette

The smaller mainland sits southwest of Panterra, closer and farther south after a further 10-degree eastward and 1-degree southward adjustment in R7. An asymmetric headland, oblique old escarpment, deeply cut southeastern coast and offset southern peninsulas replace the inverted-Australia resemblance.

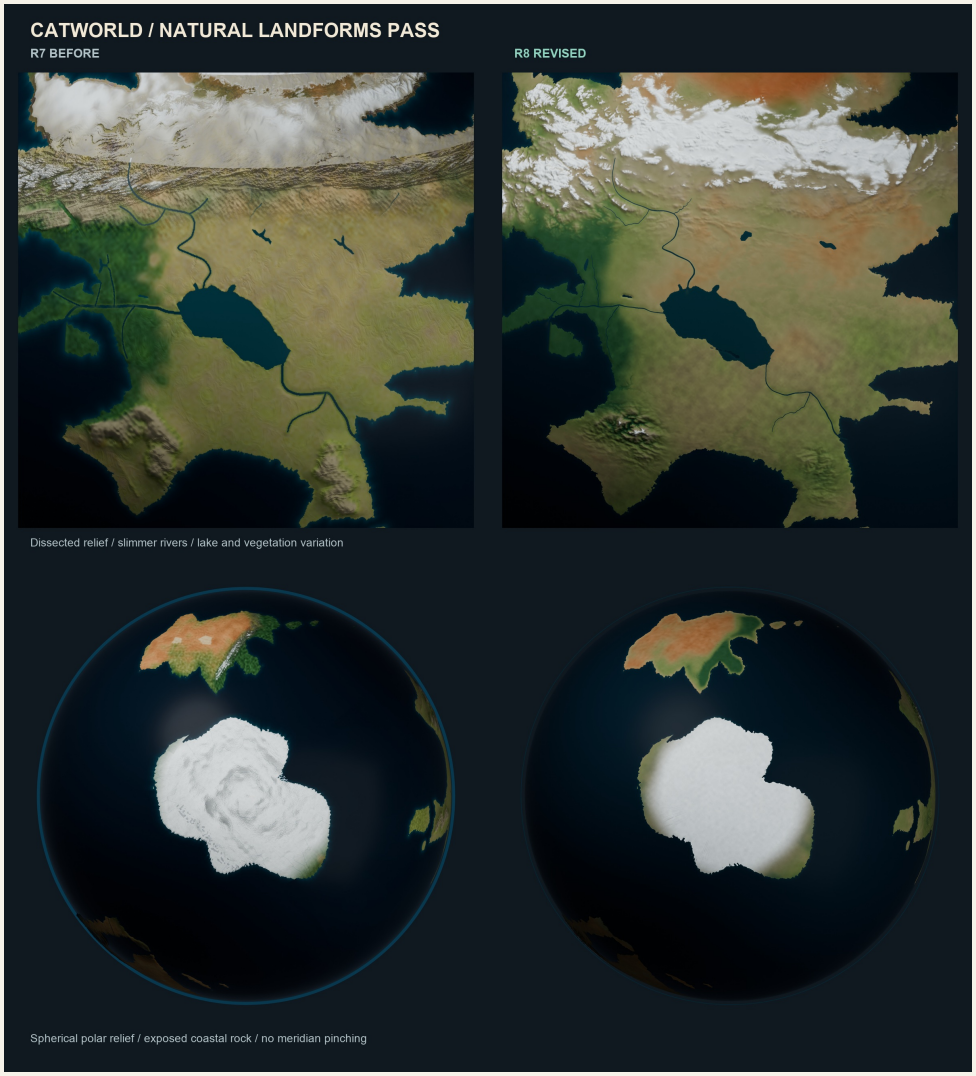
An intentionally sparse route

Five retained islands lead east toward Panterra: two large fragments, one middle island and two small remnants. The largest, irregular continental fragments cluster near the southern mainland; small, separated arc remnants pepper the approach to Panterra. Large rifted fragments recall Oceania; elongated island forms recall the Aleutians. Those are visual analogues, not copies of their real climates.

Using the globe

Front is 0 degrees, east +90, west -90 and far side 180. The southern regional camera follows the relocated mainland. Polar and regional views all render the same model.

What changed in this pass



Terrain structure

The continent layout is retained. Repeated, stretched mountain bands are replaced with more legible branching relief. Polar terrain no longer pinches where meridians meet.

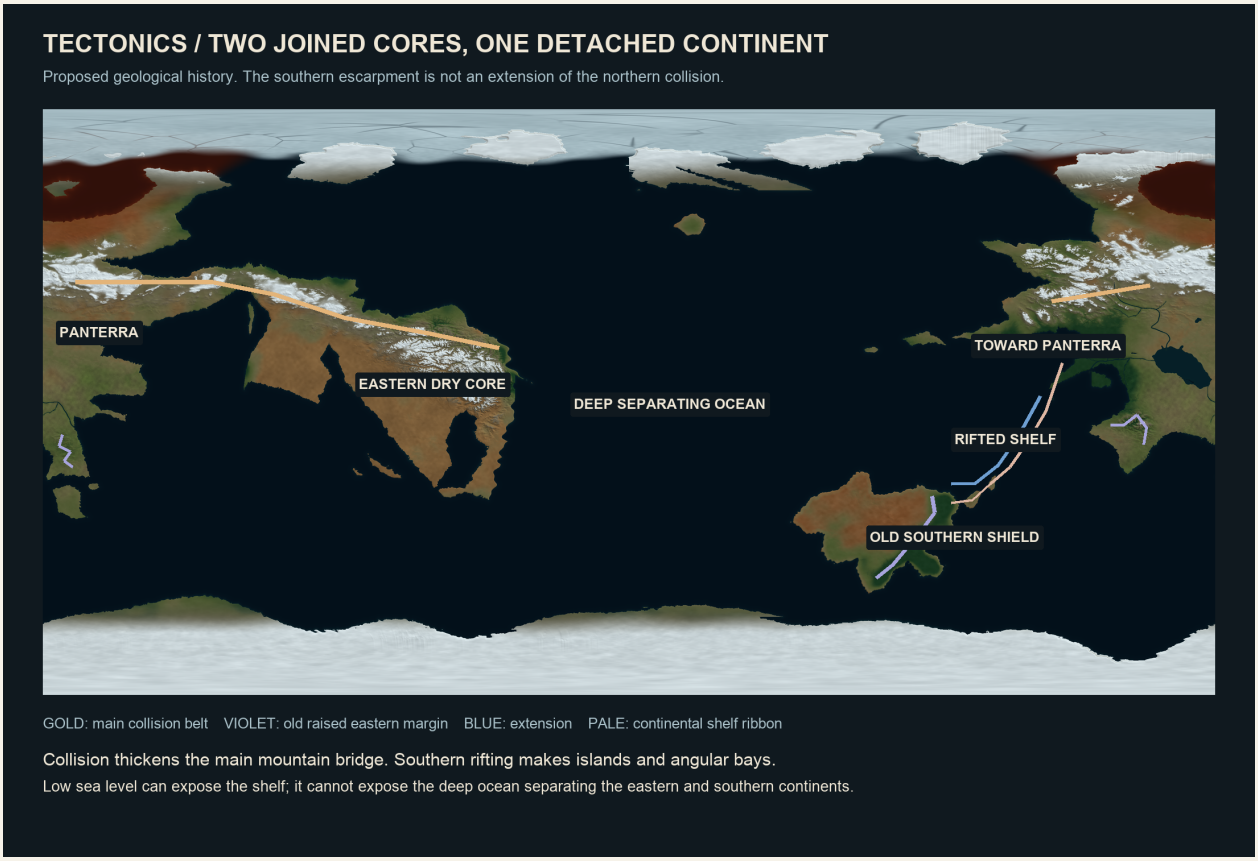
Water and surface

Rivers become slimmer and vary in width, with bends and connected tributaries. The small lakes have distinct shorelines; the reservoir remains a flooded plateau valley. Forest, steppe, farmland, exposed rock and snow share the same underlying terrain.

Closer studies

The impact water has variable mineral colour rather than a flat red fill. Broken caprock lips and near-vertical falls replace regular water ramps in the local waterfall. The feature-by-feature review is included as Terrain_Design_Review_R8.md.

A collision system and a separate old continent



Older continental crust

Panterra, the eastern dry interior and the southern mainland preserve older crustal blocks. The northern impact and its iron province predate the main collision. The southern shield has its own weathering, rifting and erosion history.

The main collision stops at the eastern margin

Panterra and the eastern block collide, thickening crust and supporting the gargantuan range. Broad folds, plateaus and valleys explain the mountain bridge. This belt does not need to continue across every continent. [S1]

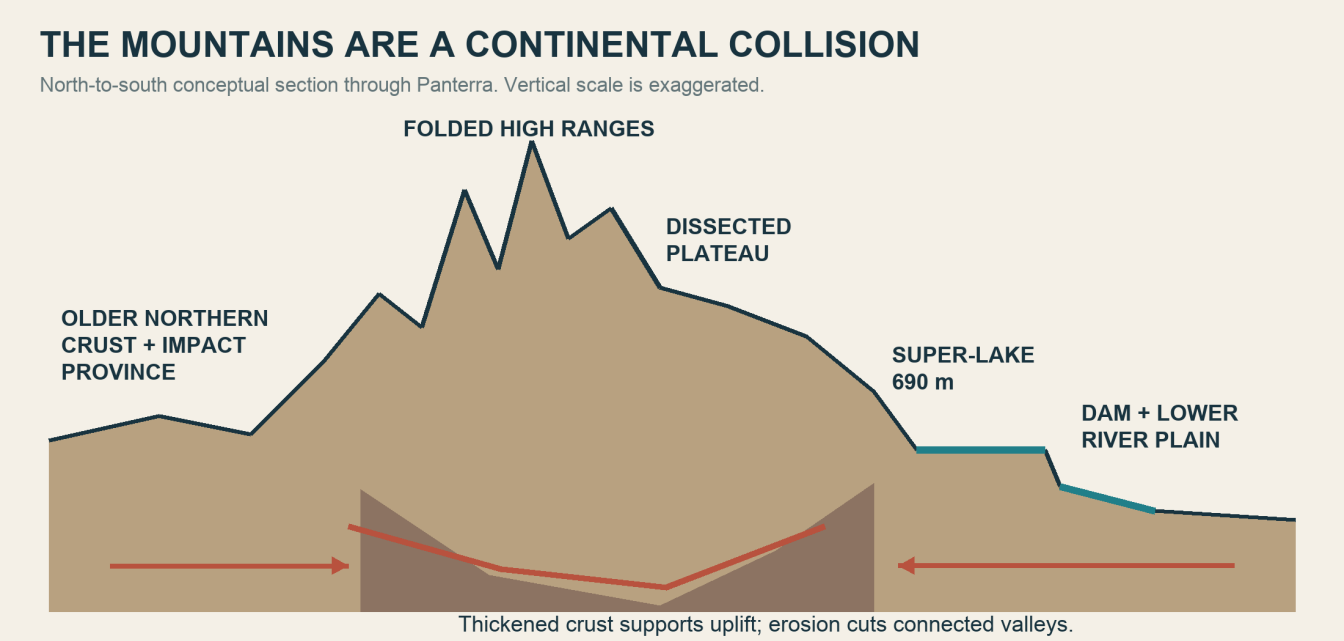
Southern rifting and an old raised margin

The independent southern continent is interpreted as an older rifted block. Its eastern escarpment is a rejuvenated inherited mountain margin, lower than the main collision belt. Extension breaks the continental shelf into large islands, gulfs and straits. This supports an Australia-like role without copying Australia.

Two different seaways

Deep ocean separates the southern continent from the eastern dry core. The long eastern shelf ribbon toward Panterra is a distinct, shallow remnant of stretched continental crust. Its former emergence is a sea-level hypothesis, not a new mountain collision. Boundary lines are schematic and plate velocities are unspecified.

Gargantuan ranges, not a row of peaks



A northern province, not one ridge

Most northern Panterran land outside the impact basin is now strongly mountainous. Multiple folded ranges and rugged massifs surround high valleys and inherited upland basins. The main crest remains the highest tier; broad supporting terrain makes it part of a continent.

A plausible mountain history

Long convergence thickens crust across a wide belt, while erosion and faulting separate individual ridges and basins. Renewed uplift can affect older northern crust without making the ancient impact responsible for every mountain. [S1]

Relief and erosion texture

A Himalayan elevation analogue supplies coherent ridge-and-valley structure at several scales. Broad foothill transitions and coastal taper prevent mountains from becoming a uniformly extruded wall. The eastern range is rotated with its host terrain.

How to draw it

Globe relief is exaggerated ten times. Draw a wide region of overlapping ranges and deep valleys, then step down into open foothills. Use the local dam and waterfall studies for true-height relationships.

Southern mountain groups

The creator's two southern mountain regions are built as broad older uplifts and a rugged peninsula spine. Eroded fold roots and later fault uplift explain high ground separated from the active northern collision. Their lower slopes feed tributaries and grade into the settled agricultural country.

Panterra: the principal story landscape



Four readable regions

A mountain-dominated north descends into steppe foothills and a dissected plateau. The great lake occupies a valley inside that plateau. Westward lie Seattle forest country and its river corridor. Southward lie fertile settlements, pasture and rolling agricultural hills.

A substantial western forest

Old growth covers a broad western watershed and inland hills, with wet ravines, logging tributaries, clearings and settlement margins. The forest gradually opens toward the plateau; it is not a thin painted coastline.

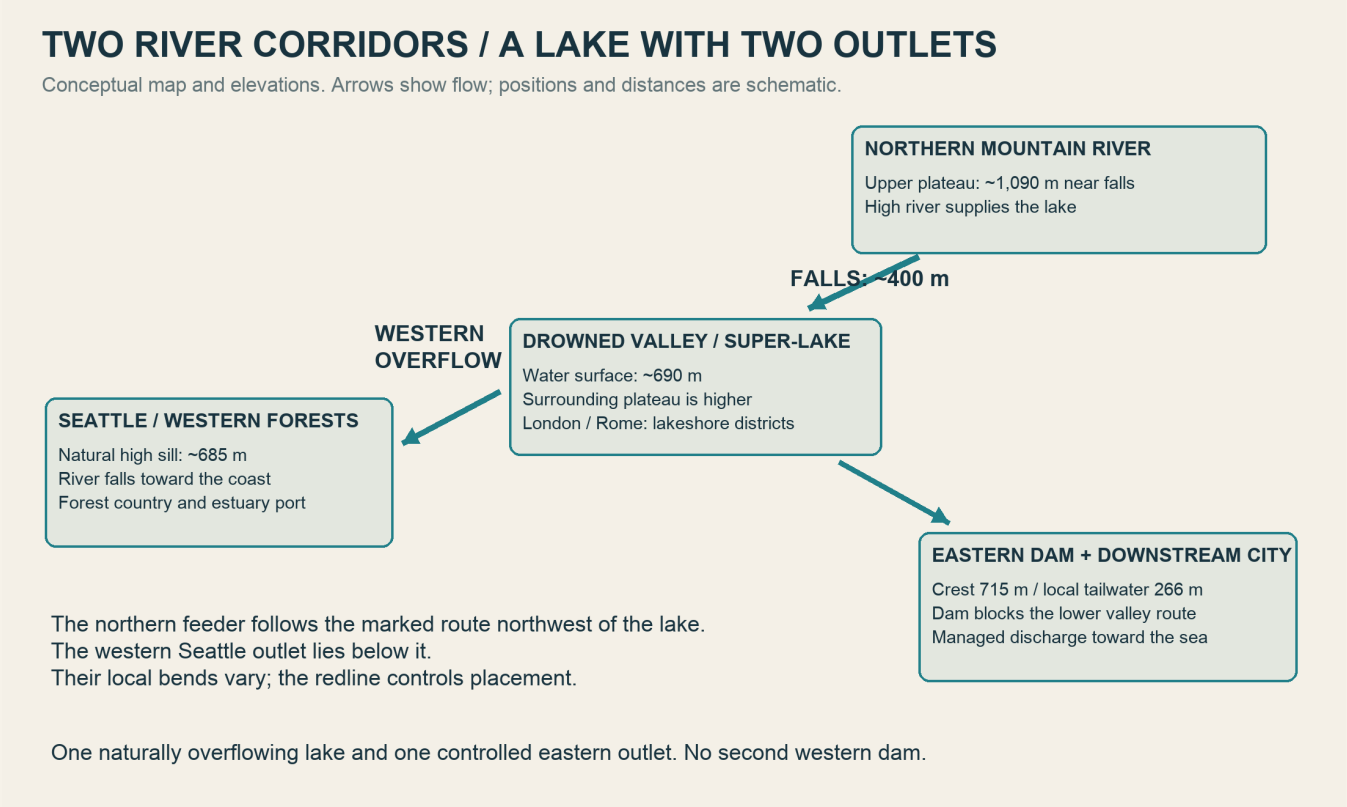
Farming and settlement

The southern green-and-gold lowlands represent agricultural country around rolling hills and the two southern mountain groups. Alluvial reaches and sheltered foothills explain settlement. High rocky interiors remain uncultivated. Individual fields and buildings are below globe resolution.

Defence and movement

The broad mountains retain lower passes and staging valleys. Rivers, escarpments and coastlines channel movement without implying that every route is easily navigable.

Two river corridors and an eastern dam



Northern feeder

Mountain runoff gathers northwest of the lake, northeast of Seattle. The main feeder follows the upper plateau and reaches the lake through a major waterfall and a single drowned valley arm. It follows the marked western plateau route instead of the crossed-out eastern route.

Western Seattle outlet

A high natural bedrock sill allows overflow from the reservoir into the western river. The river then loses elevation gradually through forest country toward Seattle and its estuary. It lies south of the northern feeder.

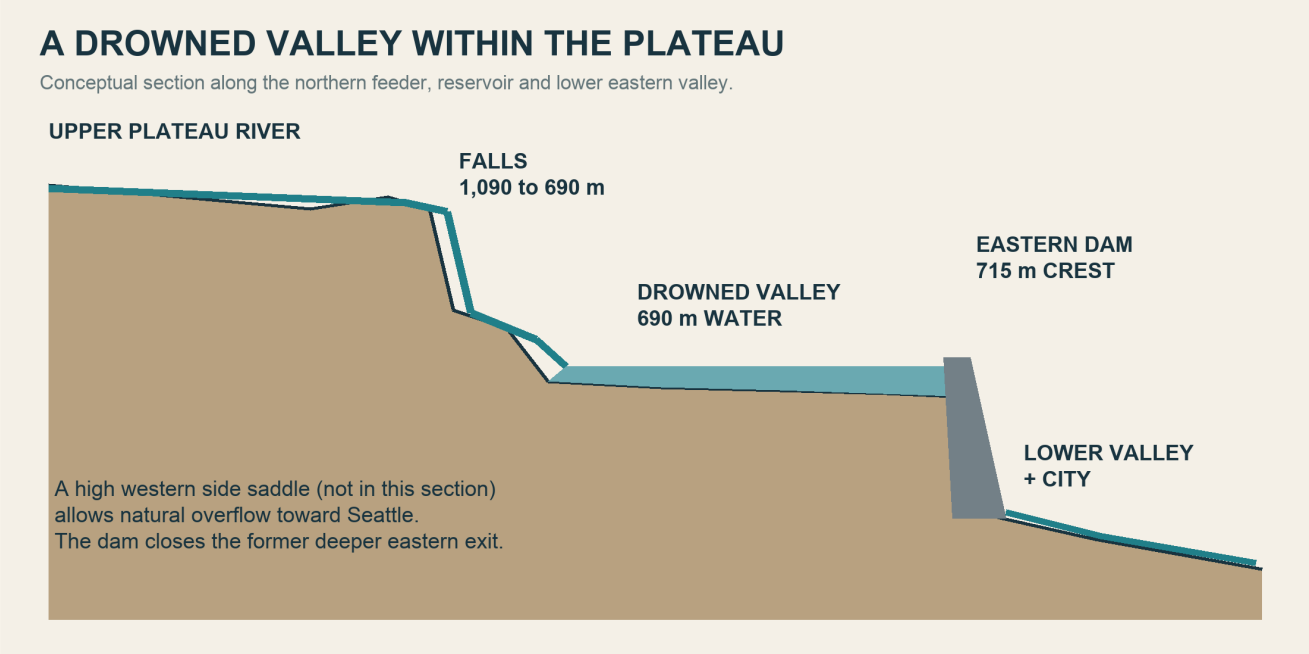
The eastern outlet remains

The megadam blocks the deeper eastern route. Controlled releases descend to the eastern downstream metropolis and sea. The reservoir therefore has a natural western overflow and a managed eastern outlet, not two incoming main rivers.

Readable watercourses

The main feeder, western outlet and eastern release dominate the globe. A few converging tributaries follow the marked western and southern catchments, with three small outlying lakes and a restrained estuary fork. Blue lines are widened for drawing clarity, not literal channel widths.

A continental lake needs a continental basin



A valley inside a plateau

The water surface is approximately 690 m above sea level. The surrounding plateau is higher, with foothills and mountains beyond. Before the dam, a smaller lake and river occupied the low valley floor; impoundment flooded connected valley branches.

Why one dam can create this lake

The eastern channel was the lower escape route. Blocking that gorge raises water until it reaches the higher western saddle. The west can then overflow naturally without draining the reservoir back to its pre-dam floor. The selected western sill is about 685 m; normal water is above it.

Why the eastern dam is lower

The downstream eastern valley floor is much lower than the surrounding plateau. The dam foundation and tailwater sit closer to sea level, while the dam crest still rises above the 690 m reservoir. A low foundation does not mean a low crest.

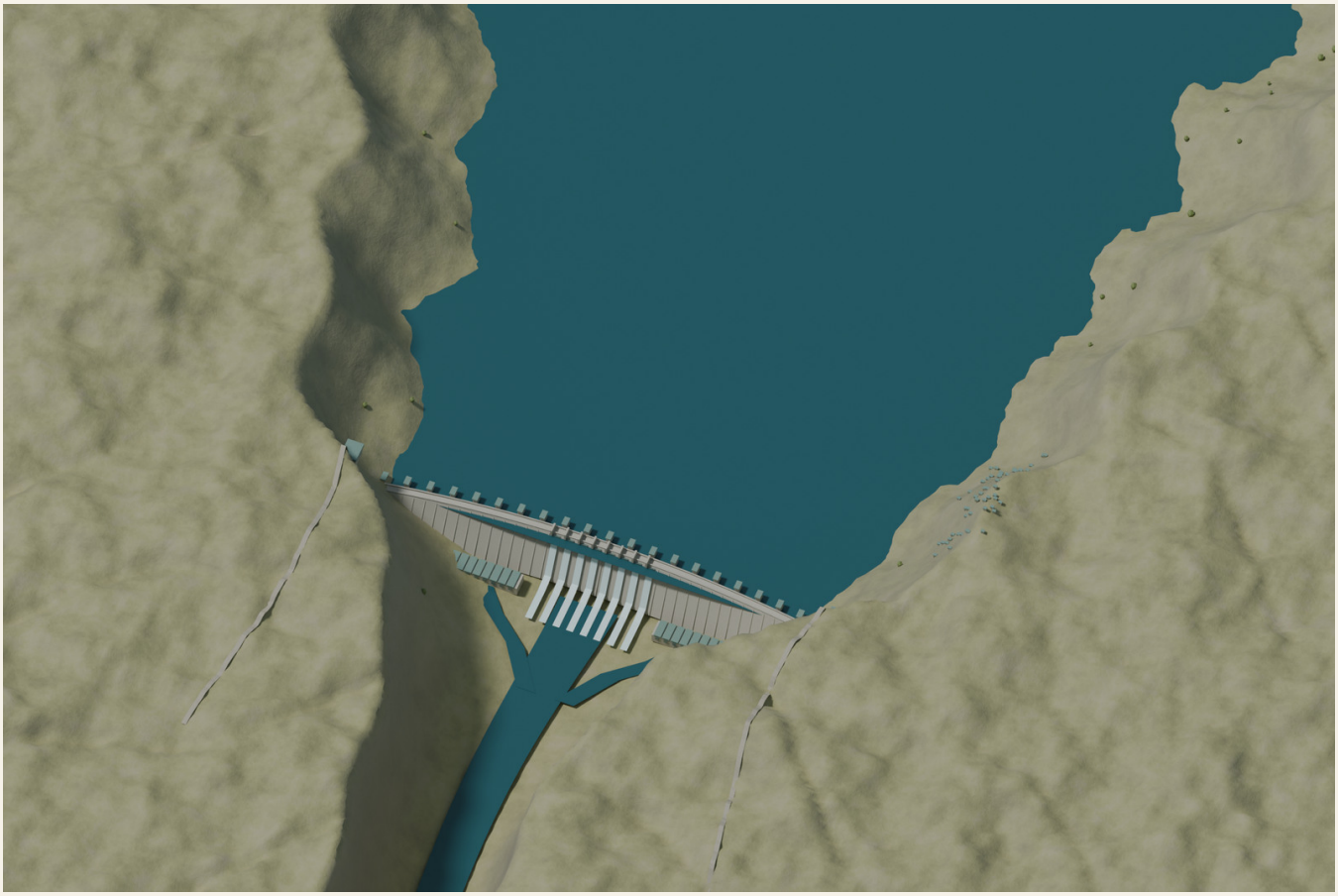
Water budget and filling time

The continental reservoir needs continental runoff and a long filling history. The northern river is the prominent feeder; smaller catchments are suppressed visually. Evaporation, western overflow and eastern releases must balance over time. The previous one-outlet discharge estimate cannot be assigned independently to both outlets.

Freshness and navigation

Throughflow can export dissolved salts, while sediment accumulates in lake arms. A broad lake or river is not automatically navigable end to end; falls, rapids, portage and seasonal operating levels need local story decisions.

The Great Panterran Dam



The designed object

The separate inset has a 4.8 km crest, approximately 470 m structural height, reservoir level at 690 m and crest near 715 m. It includes intake towers, bronze gates, spillways, turbine halls, tailraces, access roads, checkpoints and a fishing village. One inset unit equals one kilometre.

Power comes from head and flow

Using P = density x gravity x flow x head x efficiency, an assumed 10,000 m3/s, 350 m net head and 90% efficiency gives about 31 GW. A nominal 60 GW would require roughly 19,400 m3/s at that head. Lake size alone cannot guarantee that discharge. [S7]

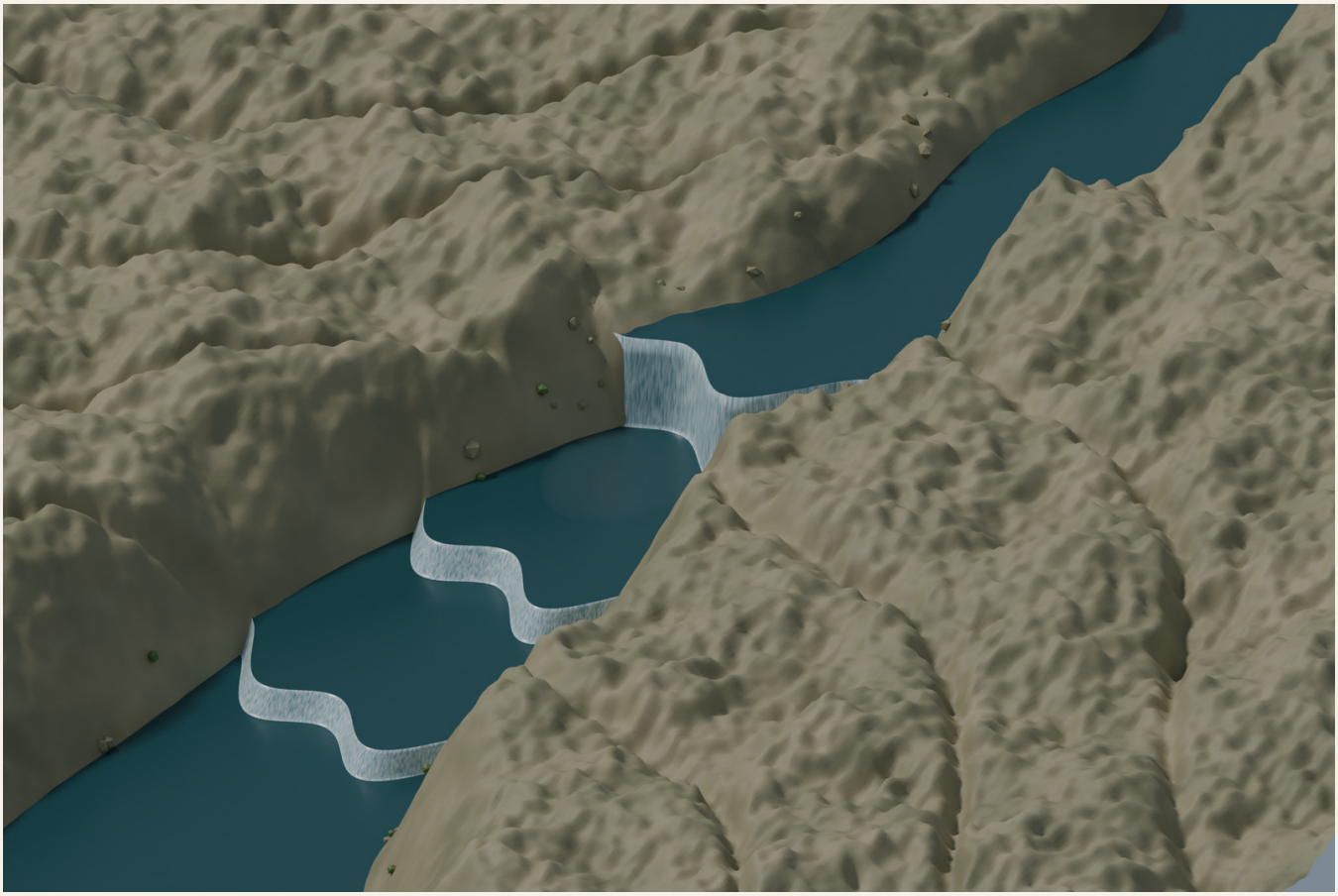
Scale and limits

The concept is larger in structural height than Hoover (about 221 m) and Three Gorges (181 m). Its illustrative 31 GW exceeds Three Gorges' 22.5 GW installed capacity. These comparisons establish ambition, not engineering feasibility. The inset is a fictional design study, not a construction-ready dam. [S8, S9]

Global and local scales

The dam wall is below the globe texture resolution. The local study shows the abrupt drop; the global river grade is smoothed. The village terrace is resolved in the inset, not in its approximate global anchor. Do not measure the local gorge from the planetary mesh.

The northern plateau waterfall



A 400 m change of level

The mountain-fed river runs across the high plateau at about 1,090 m near its end. A 250 m main waterfall reaches an 840 m plunge pool; lower cascades carry the flow down to the 690 m reservoir.

Why the waterfall is here

The river crosses a resistant bedrock lip where the upper plateau is cut by the deeper valley. Hard layers hold ledges while softer or fractured rock erodes below them. The drowned valley arm reaches the foot of the cascade.

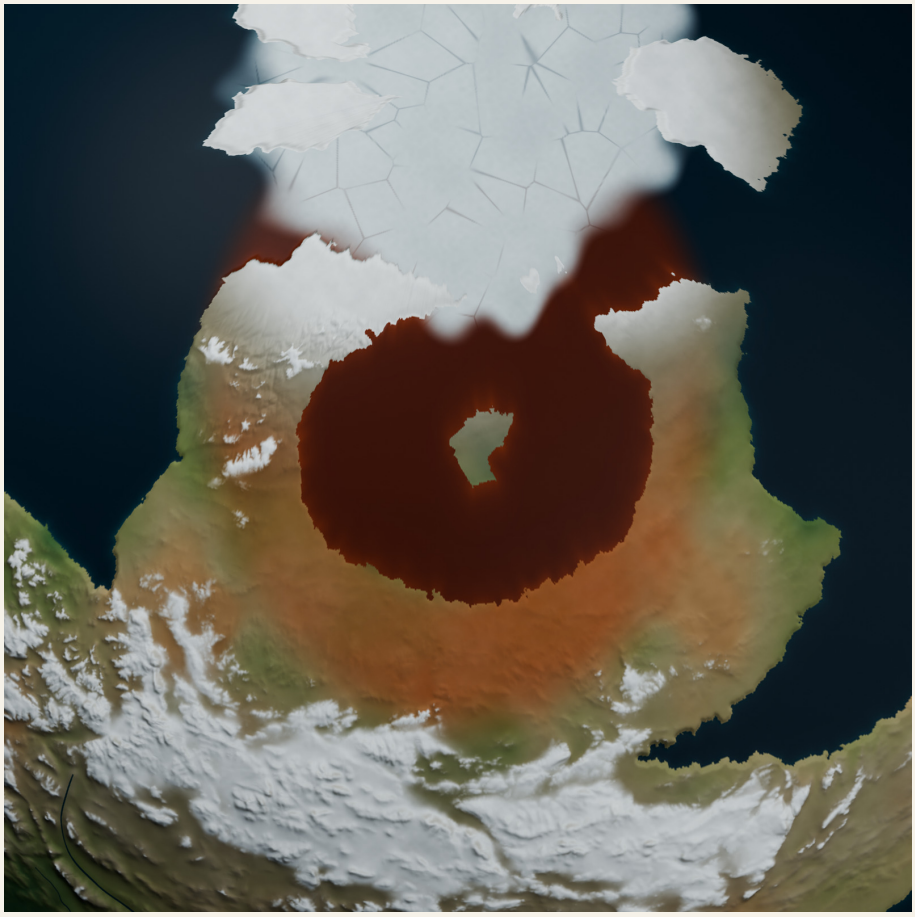
How to use the study

The local Blender model shows a continuous water surface meeting its banks, the main curtain, pools, lower cascades, steppe above and limited riparian vegetation. It is a landform reference, not a fluid simulation or a precise cutout of the coarse globe mesh.

Story implications

This is a barrier to through-navigation and a natural crossing, portage and lookout landmark. Noise, spray and seasonal discharge affect its surroundings. Its long-term erosion rate remains unspecified.

The northern impact and the red sea



An old scar incorporated into a continent

The basin has a central uplift, a broken rim and a broad apron that merges into the northern mainland. It is older than the collision belt. Erosion and subsequent tectonics modify its edges, so it reads as part of the crust rather than a circular object placed on top.

Why the red works

Rust-coloured soil and sediment suggest oxidized iron minerals such as hematite. Red water represents suspended mineral-rich material and staining, not liquid metal. Colour is a strong visual cue, but it does not prove that the iron came from a meteorite. [S3]

Ore and the story

The iron-rich northern province is a creator premise. Impact melting, fractured rock and later fluid circulation offer a proposed way to redistribute and concentrate resources. Real impact structures can host ore deposits, but rare earths and enriched uranium are not automatic products of an iron-rich impact. Keep those deck claims as separate fiction decisions. [S4; D5, D47]

Time to recover

A country-scale impact implies an ancient catastrophe. Its age must allow climate and ecosystems to recover before the current bird kingdoms. The exact age, original diameter and ore inventory remain unspecified.

The dry eastern core



A southward swing

The eastern block is rotated eight degrees clockwise around its junction with Panterra. The far end is more vertical in the world view, while the connecting mountain belt remains continuous.

Relief and rain shadow

The enlarged northern mountain margin gives way to dry lee-side country with gravel basins, dunes, salt pans and dissected escarpments. The west/left dry side from the creator redline remains dry. Coastal pockets can still receive rain.

Coasts with a history

The long rift gulf, irregular peninsulas and exposed faulted margins move with the continent. Rivers and geographic anchors are transformed with the land, avoiding detached overlays.

Open ocean beyond

There is no bridge or island arc between this continent and the southern mainland. Their geology and migration roles are distinct.

The independent southern continent



An independent southern world

The redesigned shield remains entirely south of the equator. Its angular northwestern headland, asymmetric southeast embayment and offset southern peninsulas give it a new outline rather than a rotated Earth analogue.

Dry interior, wet margins

Old weathered uplands, stony basins and salt pans occupy the interior. The eastern escarpment catches moisture; the cooler southern coast has wetter inlets and slopes. These climatic contrasts persist during southern spring.

Closer, but still separate

The southeastward adjustment shortens the passage toward Panterra and puts the mainland more clearly to its southwest. Water still clearly separates the modern mainland and the island chain. There is no new land neck.

Old homeland

Its identity as the cats' old homeland is still a proposal. The sheltered eastern shelf, coastal resources and former crossing can support that history without fixing political borders or final place names.

The eastern archipelago and the old crossing



A sparse curved archipelago

Five islands replace the crowded belt; the southern mainland is closer, shortening the crossing. Larger fault-bounded fragments transition into slender volcanic-looking islands, separated by conspicuous water gaps. The chain runs east toward Panterra only. The Aleutians inform its arc curvature and aligned peaks, not the depth of its crossing shelf. [S14]

What could form a former bridge

Submerged continental saddles provide the possible old crossing. The volcanic-looking peaks alone do not imply shallow water: the model separately includes a continental shelf beneath the route. At the illustrative 65 m lower sea level, that designed shelf emerges.

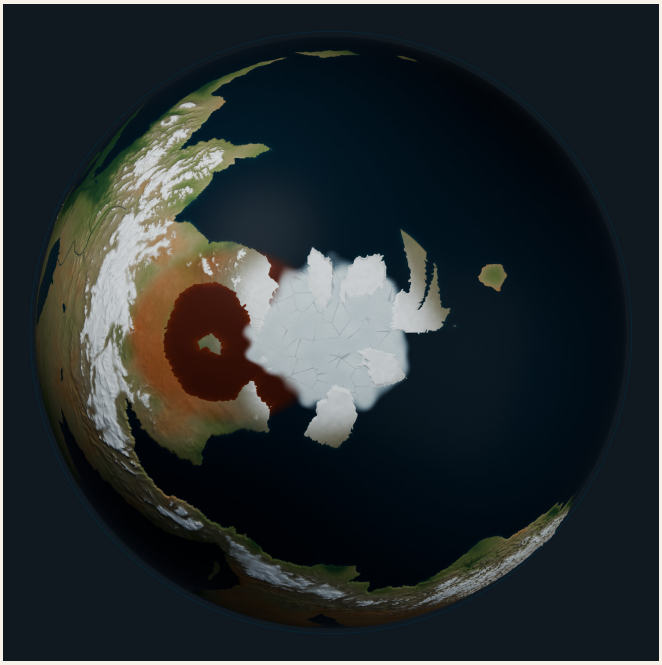
Ice and sea level

Net growth of grounded ice can lower the ocean. Floating sea ice alone is insufficient. Strong southern winter could support the crossing only if added southern land ice outweighed losses elsewhere. This remains a hypothesis. [S11]

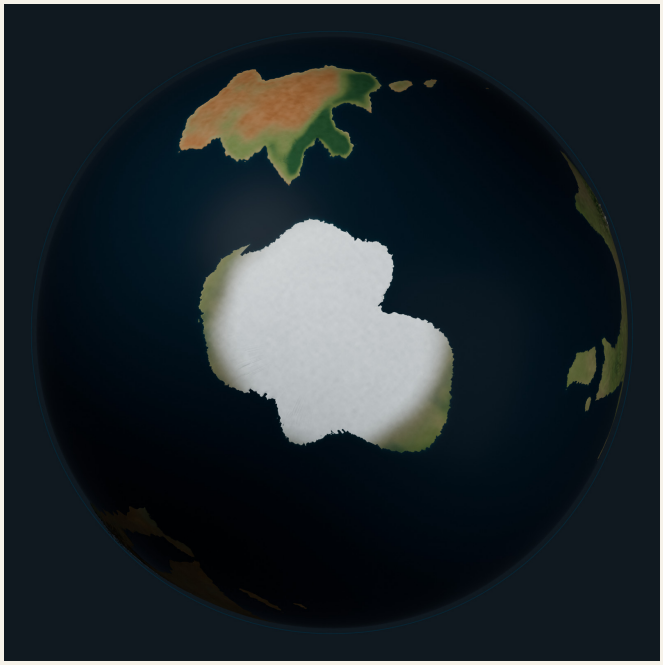
Limits

Sparse islands do not prove that the whole gap was once walkable. The lowstand test verifies designed connectivity, not real ice volume, dates, currents, or travel safety. The global grid cannot specify every historical strait.

Two poles with different identities



NORTH / OCEAN + ISLANDS



SOUTH / ICE CONTINENT

A darker northern winter frontier

With 35-degree tilt, the polar circles lie at 55 degrees latitude. At the chosen late-autumn epoch, the Blood Lake district is already in geometric polar night. Reference cameras remain lit for drawing; they do not represent the actual sun position.

Northern ice and relief

Snow covers high ranges and spreads across northern uplands. Ocean pack and broken shore ice extend around the polar margins. The iron-red lake remains a readable design landmark; the exact amount of unfrozen water needs a thermal and salinity explanation.

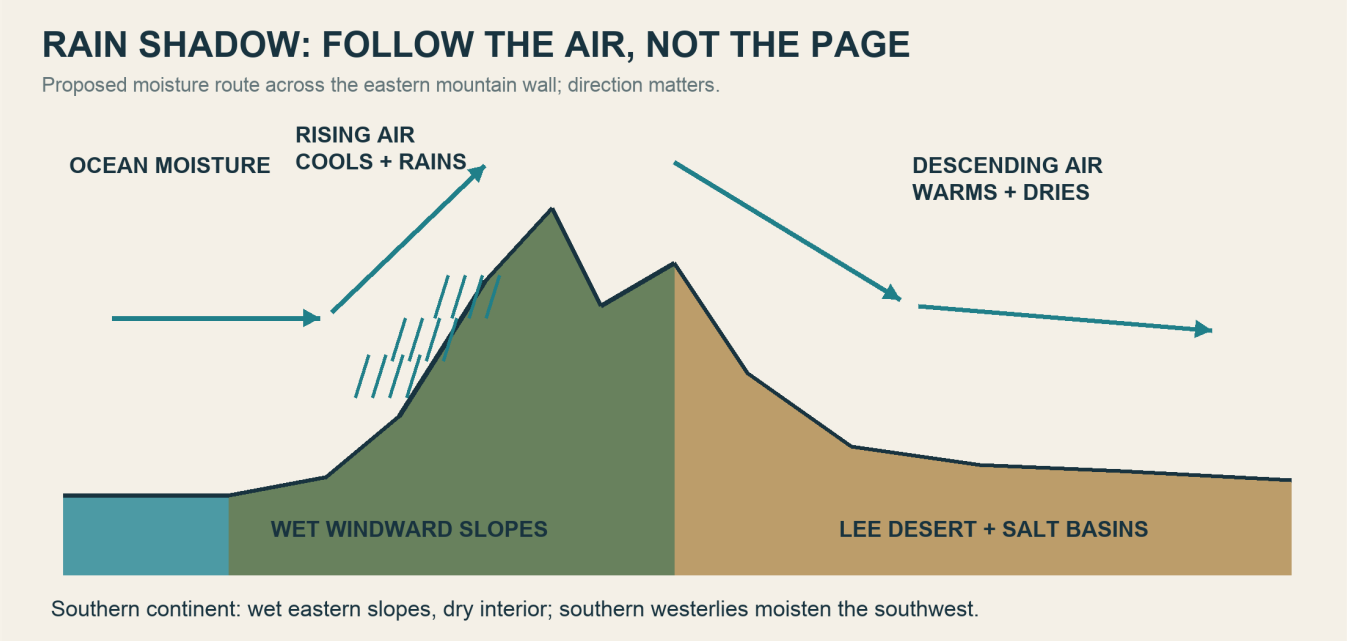
Southern spring

Grounded polar ice and mountain glaciers remain while coastal seas reopen. More extreme tilt also brings stronger summers, so a permanent ice map cannot be inferred from winter severity alone.

Ice storage and migration

The large southern polar landmass can store grounded ice. Whether its long winter actually produced the proposed lowstand requires a global mass balance. Opposite seasons by themselves do not establish the crossing.

Climate follows terrain and exposure



Southern continent: wet eastern slopes, dry interior; southern westerlies moisten the southwest.

Western moisture and old growth

The western country is the main forest province. Maritime moisture, wet ravines and sheltered lower slopes support dense stands around Seattle. Forest coverage fades gradually inland rather than ending at a rectangular boundary.

Northern foothill steppe

Open grassland and scrub occupy the northern foothills below the high mountains and the drier plateau interior. Water follows valleys; narrow riparian vegetation does not turn the whole region into forest.

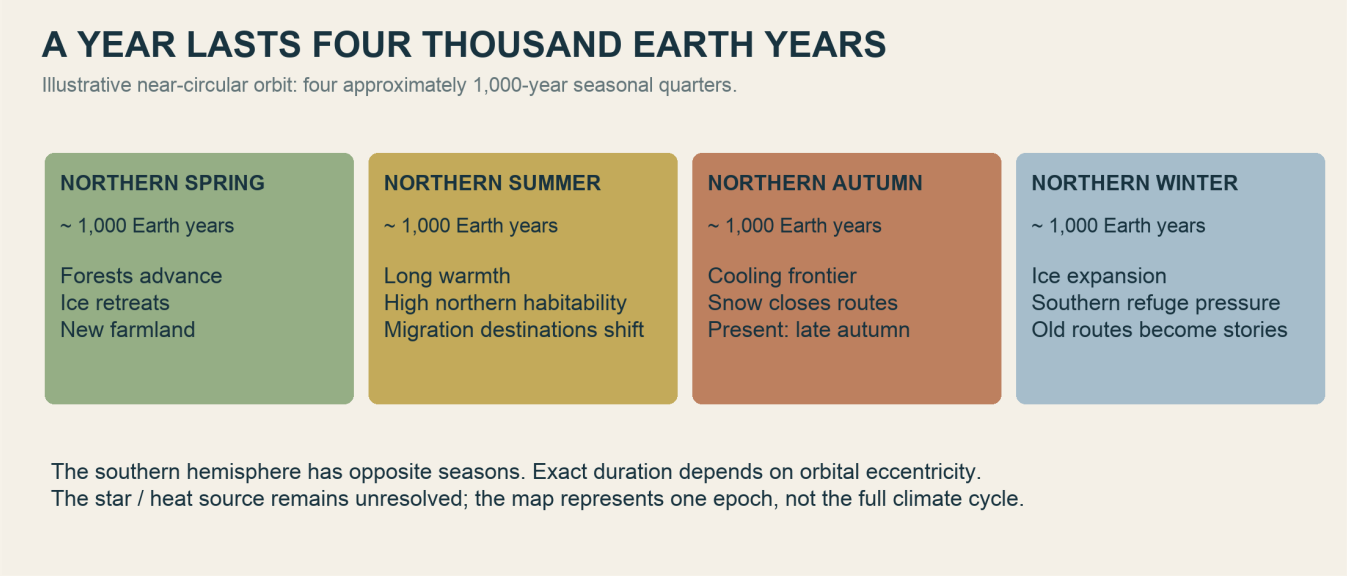
Fertile southern country

The south and major settled districts have agricultural mosaics, pasture and rolling hills. Lake margins and river terraces support crops and transport. Soil fertility is a design premise, not a calculated agronomic yield.

Rain shadows and model limits

Wet windward slopes and dry lee interiors follow mountain exposure and proposed winds. Seasonal rain belts and currents are schematic. Regional colours combine terrain and climate reasoning but are not a circulation-model result. [S2]

Living inside thousand-year seasons



An explicit working date

Use solar longitude 255 degrees: about 833 years after the northern autumn equinox and 167 years before northern winter solstice in a circular 4,000-year orbit. The south is the same distance through spring, approaching summer. These precise numbers are proposed for continuity; the seasonal pairing follows the creator's direction. [S12]

A stronger tilt

The adopted tilt is 35 degrees. Higher tilt increases summer sunlight and reduces winter sunlight at high latitudes, strengthening seasonal contrasts. The thousand-year duration allows long ecological and human responses, but does not mean temperature declines without limit for a thousand years. [S13]

Weather does not wait for the boundary

Northern high latitudes can already feel wintry during late autumn. Southern uplands and the polar ice sheet can remain cold during late spring. Latitude, elevation, oceans and inherited ice modify local conditions. A thousand-year season is a changing climate interval, not one daily weather pattern repeated unchanged.

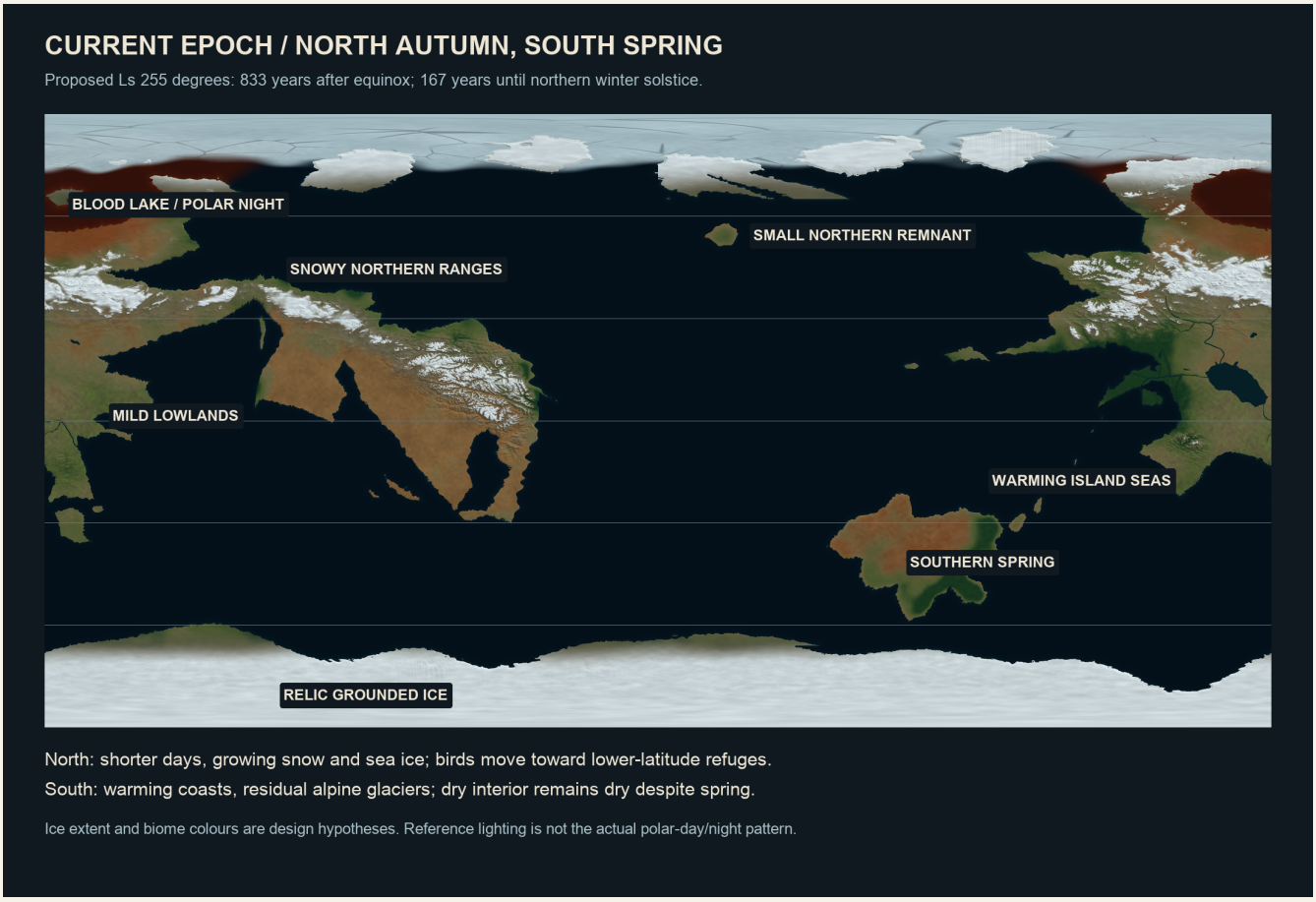
Earlier crossing and current bird movement

A proposed crossing 1,100 years ago falls within the previous southern winter / northern summer. A lower ocean exposes the shelf while northern destinations are warmer. Today, net southern ice loss has drowned that corridor, while advancing northern cold pushes bird communities south. Exact ice history and migration dates remain to be agreed.

The heat source is still unresolved

A 4,000-year orbit around a one-solar-mass star has semimajor axis about 252 AU. A Sun-like star would not provide Earth-like warmth there. The setting needs a different stellar/orbital or additional-heating premise. The seasonal visuals assume that a habitable thermal solution exists. [S6]

The present seasonal distribution



One stated epoch

Solar longitude remains 255 degrees in the proposed near-circular orbit: about 833 Earth years after the northern autumn equinox and 167 before northern winter solstice. The south is in late spring.

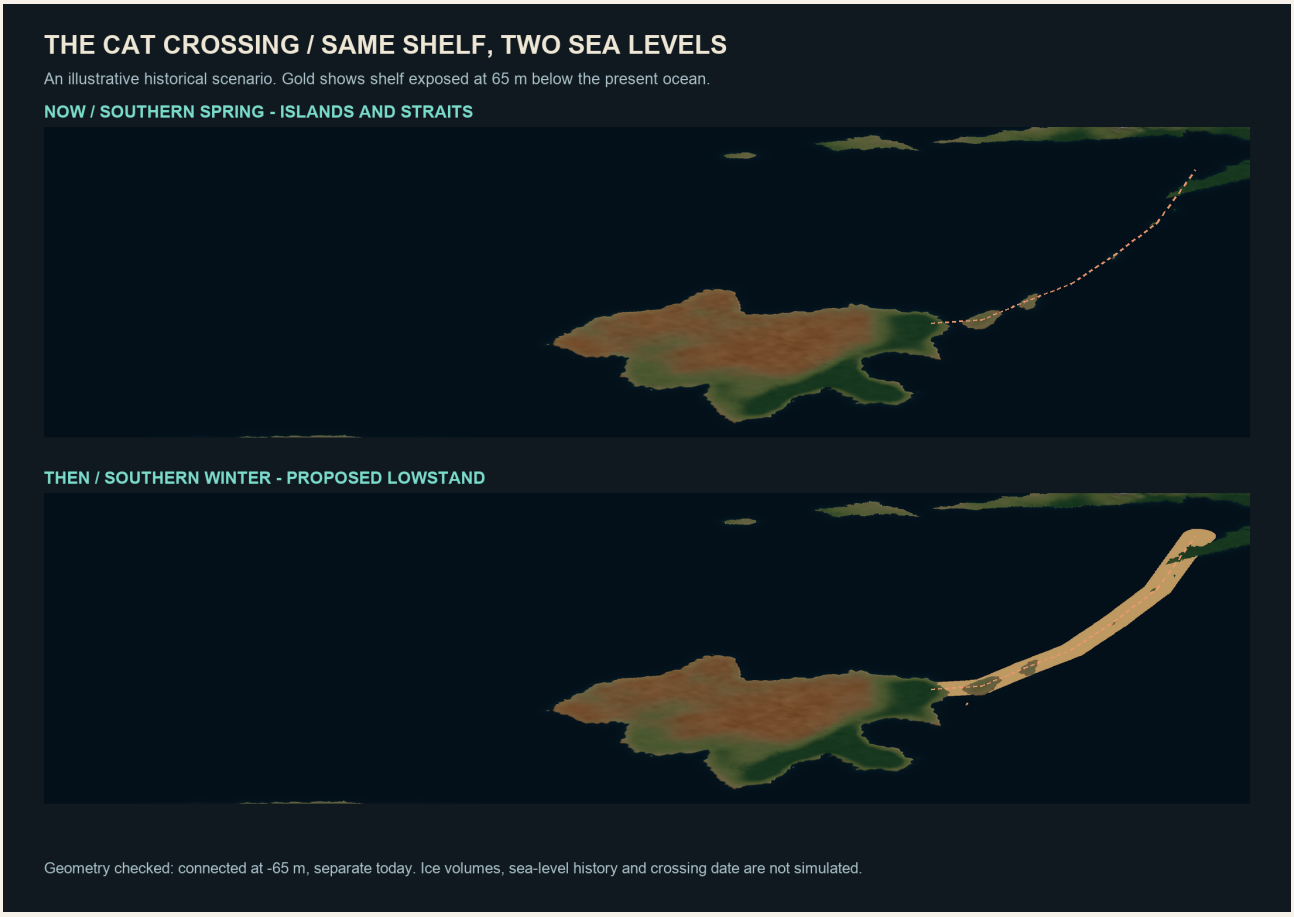
Daylight is recalculated

The model now derives daylight from 35-degree tilt and the stored epoch. The Blood Lake has zero geometric daylight at this snapshot. The southern mainland has long spring days. The delivered CSV gives region-specific values without atmospheric refraction or terrain horizons.

Surface appearance versus illumination

Snow, vegetation and sea ice show a proposed seasonal state. Broad daylight lights are used in every reference camera so the geography remains visible. The guide separates that artistic lighting from the planet's actual day/night geometry.

The cat crossing: then and now



Then: southern winter / northern summer

The lower panel lowers the ocean by 65 m in the stored elevation field. Exposed shelf links the old southern mainland to Panterra. The route can support generations of settlement and movement, rather than requiring one uninterrupted march. The 1,100-year date places it within the previous southern winter.

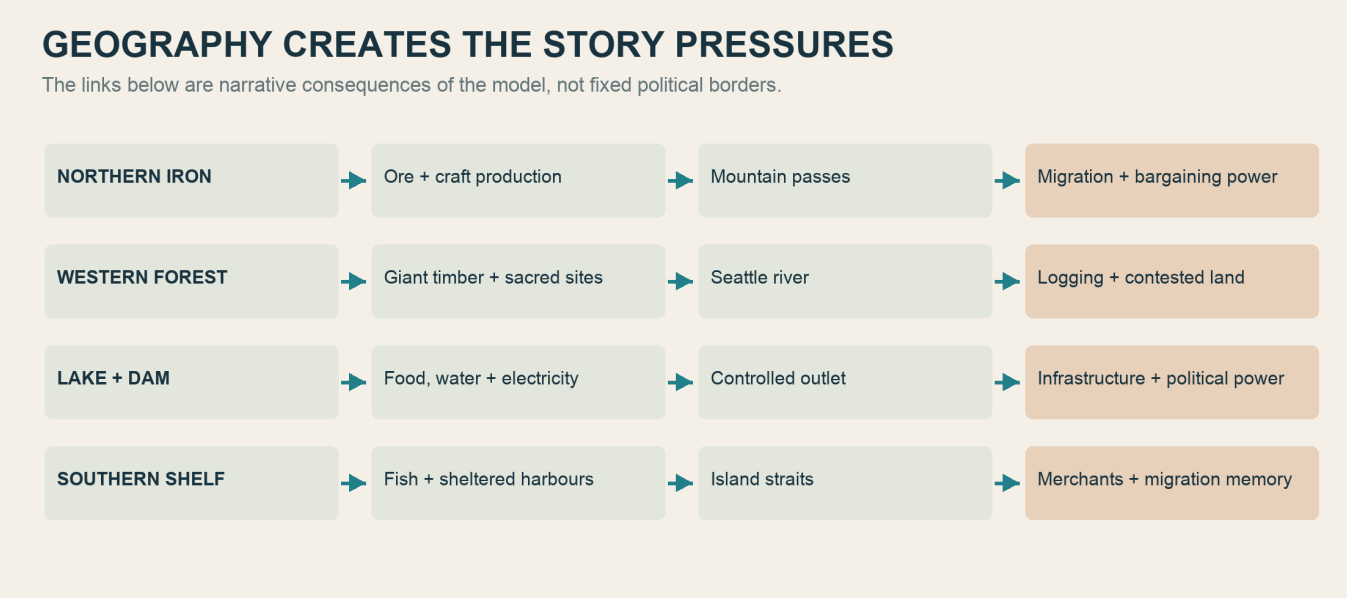
Now: southern spring / northern autumn

The same shelf is drowned and the southern mainland is separate. Islands retain traces of the former route. Northern birds now face cooling homelands and move toward milder Panterra. The present archipelago is a shipping landscape, not a modern dry crossing.

What the check proves

A periodic-grid connectivity test confirms the designed lowstand geometry. It does not establish the required global ice volume, exact water depth in each local channel, bridge longevity or the historical migration itself. Those remain stated worldbuilding assumptions.

Resources create routes and conflicts



The material contrast

The deck's Panterran economy relies on bronze/brass, scarce iron, waterways and electrical metallurgy. Northern birds bring access to an iron-rich province. The map supports that contrast through separate resource districts and costly passages between them. It does not require every useful metal to have the same geological origin. [D47]

The forest is contested infrastructure

Giant trees supply timber and support sacred avian sites. Rivers make those forests economically accessible to logging interests. Deforestation, damaged spawning streams and lost nesting places can therefore be local, visible causes of conflict rather than abstract background. [D4, D5]

Two different kinds of water control

The eastern dam controls releases, power and part of the lake-level budget. The natural western overflow responds to water level rather than a gate. Seattle depends on that western river, so lake management affects communities on both sides of the plateau.

Limits of this atlas

Ore tonnages, forestry yields, crop output, electrical demand and transport capacity are not quantified. The map suggests where these systems interact; it is not a balanced economy simulation. Political ownership and final settlement names remain creator choices.

Cities and settlements need reasons

Seattle / forest country and lower-river port

Seattle names a substantial western region as well as its port. Old-growth woods, working forest, tributary settlements and fertile clearings surround the downstream river. Place the core settlement on raised ground near the estuary, with wharves on sheltered reaches. This is no longer the upstream timber-city layout.

London and Rome / lake country

London occupies southwestern lakeshore country and Rome a high eastern bank. Both can combine fisheries, lake shipping and agricultural hinterlands. Shorelines need harbours, terraces and seasonal operating levels before detailed city plans are drawn.

Amsterdam / downstream outlet

The large downstream city is a trade and processing center with canal links and alluvial land. Its wealth can come from controlling exchange and infrastructure rather than possessing rich local ore. This follows the deck's trade-outlet concept. [D41]

Milan / merchant island

The southern island is a useful site for maritime customs, resupply and finance. Its precise role in the inland supply chain still needs chosen routes. Do not assume the lake and sea share one uninterrupted navigable level.

Small places

Fishing villages belong in sheltered coves and on stable terraces. Crossing towns belong at ferries, bridges and junctions. Checkpoints belong at supply bottlenecks. Ancient ruins should predate the modern routes they overlook. The dam inset shows one village proposal; the planetary model does not contain a city-building kit.

Travel continuity

At 6,100 km radius, one degree of latitude is about 106 km. Several degrees between markers can mean weeks of travel, depending on transport and terrain. Route distances, seasonal pass closures and communication speeds must be set before simultaneous episode journeys are scripted.

Migration, sacred trees and choke points

A northern departure, not a featureless arrow

The bird homeland occupies the impact district and surrounding northern country in the draft. Cooling affects nesting sites, food supplies and accessible routes. Families, livestock, freight and institutions do not move like a single airborne flock. [D2, D5]

Why the passes matter to birds

Flying individuals can cross barriers that stop carts, but high altitude, weather, loads, feeding stops and ground-based supply still create preferred corridors. Western, Central and Eastern Gate are proposed lower passages. They are logistical choke points, not an absolute biological wall.

Where control becomes visible

Checkpoint stations can cluster at valley mouths, crossings, staging fields and supply depots. Beyond them, routes branch toward timber country, lakeshore towns and the downstream metropolis. The model locates candidate gates; it does not specify a border regime or a completed road network.

The forest preserves memory

Ancient avian carvings and cliff ruins sit above the western forest corridor. The old-growth refuge provides a setting where material evidence, oral history and present land claims meet. Logging can expose ruins while destroying the living landscape around them. Those are narrative proposals grounded in the deck's sacred-tree material. [D5, D9]

Historical continuity

The latest brief sets two distinct migrations: an older cat crossing enabled by southern winter, and current birds moving south as northern winter approaches. The 1,100-year crossing date is illustrative. Earlier deck dates, including roughly 700 years of northern occupation, need reconciliation with the chosen epoch; they are not silently treated as confirmed history. [D2, D5, D45]

Use the globe as a drawing reference

Choose the angle before drawing

Open the offline atlas and select the hemisphere or regional view. Orthographic cameras show the actual sphere without a map-projection stretch. Land near the limb is foreshortened; choose a closer-centered view when a region is the drawing subject. The annotated versions locate features on the same renders.

Open the editable models

The globe contains thirteen saved cameras with matching lights, packed textures and named anchors. The dam has two reference views. The waterfall has its own editable local study, with one kilometre per Blender unit and true vertical scale.

Resolution and exaggeration

An 8192 x 4096 colour map and three 4096 x 2048 physical maps support the globe. Thirteen globe views, two dam views and one waterfall view make sixteen clean and sixteen annotated references. The globe uses 10x relief and widened rivers; local studies use true relief. Small buildings, individual trees and engineering details remain below global resolution.

What can be edited

Geographic anchors, directed river paths and planet parameters are stored in source JSON. The build pipeline reconstructs earlier terrain stages before applying R6, R7 and the R8 terrain pass, so transforms are not accumulated by repeatedly rotating a finished model. The packed Blender files remain independently editable.

Light and atmosphere

Clouds are omitted to expose the terrain. Each camera has broad daylight from a useful drawing angle, not one simultaneous astronomical illumination state. The blue limb is a restrained visual atmosphere; it is not an atmospheric scattering simulation.

Decisions for the creator to make next

1 / Heat source and seasonal physics

Keep the adopted 35-degree tilt and northern-autumn/southern-spring direction. Decide the habitable heat source and validate the precise epoch, temperature distribution, ice mass and sea-level history.

2 / Homeland and coastline approval

Review the separate southern mainland and eastern archipelago, and decide whether this is officially Old Panterra. Panterra and the eastern continent remain mountain-connected. Final names, factions and political borders remain open.

3 / Reservoir and waterfall history

Develop the pre-dam valley, filling timescale, natural western sill, eastern operating rules and the waterfall's resistant bedrock. The local fall is a proposal with a 250 m main drop and 150 m of lower cascades.

4 / Energy and material limits

Choose reliable rather than nominal power output, catchment assumptions, industrial demand and the role of scarce iron. Treat naturally enriched uranium and any unusual cooling mechanism as separate fictional premises requiring decisions.

5 / Ice budget, crossing and travel

Approve or revise the proposed 65 m lowstand and 1,100-year crossing date. Test net grounded-ice storage, relative sea level and local shelf elevations before declaring a safe historic route. Fix journey distances and seasonal pass accessibility for scripts.

6 / Local production detail

Develop Seattle forest districts, agricultural settlements, lake shipping, eastern dam operations and the northern passes at regional scale. Use the broad continental atlas for continuity; commission local maps for individual journeys and scenes.

R8 / A more natural terrain language



Relief and drainage

Repeated mirrored relief is replaced by bounded elevation sampling and broader regional terrain. Slender river corridors have variable widths, bends, graded valleys and joined tributaries. The main lake remains a level drowned plateau valley; three small lakes have distinct asymmetric shorelines.

Land cover and ice

Forest, steppe, agricultural lowlands and bare rock blend through moisture and altitude fields. Snow responds to elevation and exposed ridges. Polar relief uses spherical coordinates to avoid pinching at the pole; cold shores show rock and tundra rather than a green halo.

Basin and local studies

The impact rim has uneven eroded sectors and patchy iron staining. Sediment colours vary near water margins and river mouths. The waterfall study has broken caprock lips, deeper plunge pools, channel bends and talus; the dam gorge has irregular eroded shoulders.

Continuity and limits

The approved continents, five-island crossing and main river corridors remain. These are process-informed design choices, not a simulated drainage, erosion, ice or climate solution. River widths and globe relief remain exaggerated for drawing reference.

Sources and reference notes

D / Project references

CATWORLD PITCH DECK.pdf, 50-page development draft, and map sketches 3-5. D2 = overview/migration; D4 = Panterra; D5 = Flock; D9 = Cyrus context; D12 = Pieter/dam village; D38-42 = map concepts; D45 = chronology; D47 = materials and dam. Page numbers refer to PDF pages. The latest creator corrections take precedence. Private message screenshots and the deck itself are not redistributed in this package.

S1 / Collision mountains

USGS, The Himalayas, https://pubs.usgs.gov/gip/dynamic/himalaya.html ; National Park Service, Collisional Mountain Ranges, https://www.nps.gov/subjects/geology/plate-tectonics-collisional-mountain-ranges.htm

S2 / Rain shadows

NOAA, Teek and Tom educator guide, lesson 2, https://oceantoday.noaa.gov/teekandtom/educators-guide/lesson-2.pdf

S3 / Red iron oxides

USGS, Find-A-Feature: Red Rocks, https://www.usgs.gov/educational-resources/find-feature-red-rocks

S4 / Impact-related resources

Reimold and colleagues, Economic mineral deposits in impact structures: A review (2005), abstract record at https://hero.epa.gov/reference/1435484/ . The resource mechanisms are analogues, not a measured ore model for Terra.

S5 / Polar analogues

National Snow and Ice Data Center: https://nsidc.org/learn/parts-cryosphere/sea-ice and https://nsidc.org/learn/parts-cryosphere/ice-sheets

S6 / Orbital period

NASA, Orbits and Kepler's Laws, https://science.nasa.gov/solar-system/orbits-and-keplers-laws/ . The 252 AU figure is this guide's calculation for a one-solar-mass primary.

Sources and reference notes / continued

S7 / Hydropower

US Department of Energy, How Hydropower Works, https://www.energy.gov/cmei/water/how-hydropower-works . All Terra discharge, head, efficiency and power values are explicitly assumed examples.

S8-S9 / Dam scale comparisons

US Bureau of Reclamation, Hoover Dam, https://www.usbr.gov/projects/index.php?id=122 ; China Energy Engineering Corporation project profile, Three Gorges maximum height and capacity, https://en.ceec.net.cn/attach/0/1903261138048558920.pdf

S10 / Warming and aerosols

NASA, Causes of Climate Change, https://science.nasa.gov/climate-change/causes/ ; NASA, Aerosols: Small Particles with Big Climate Effects, https://science.nasa.gov/science-research/earth-science/climate-science/aerosols-small-particles-with-big-climate-effects/

T / Terrain provenance

Mapzen Terrain Tiles on AWS, with USGS SRTM/GMTED elevation sources. A Himalayan mosaic was warped, rescaled and combined with fictional terrain. https://registry.opendata.aws/terrain-tiles/ . Full provider attribution is bundled in source/TERRAIN_ATTRIBUTION.md. No provider endorses this fictional model. Drainage reference: USDA NRCS Soil Survey Manual, chapter 2, https://www.nrcs.usda.gov/sites/default/files/2022-09/SSM-ch2.pdf ; sediment plumes: NASA, https://svs.gsfc.nasa.gov/15034 .

S11 / Ice and sea level

NASA, Melting sea ice versus land ice: https://sealevel.nasa.gov/faq/23/whats-the-difference-between-melting-sea-ice-and-melting-land-ice-on-sea-levels/ ; NOAA, Local versus global sea level: https://oceanservice.noaa.gov/facts/sealevel-global-local.html . These support the mechanisms, not the fictional 65 m or 1,100-year values.

S12 / Opposite seasons

NOAA / National Weather Service, Why Do We Have The Seasons? https://www.weather.gov/fgz/Seasons . The 255-degree epoch and dates are this guide's calculations for a circular 4,000-year orbit.

S13 / Stronger axial tilt

NASA Science, Milankovitch (Orbital) Cycles and Their Role in Earth's Climate. https://science.nasa.gov/science-research/earth-science/milankovitch-orbital-cycles-and-their-role-in-earths-climate/ . General tilt-season mechanism; the 35-degree alien-world climate is a design extrapolation.

S14 / Island-arc analogy

USGS, Seismicity of the Earth 1900-2010, Aleutian arc and vicinity. https://pubs.usgs.gov/publication/ofr20101083B . The real Aleutians are a subduction arc. The model combines an old arc with a retained continental shelf; a deep volcanic arc alone would not be exposed by a 65 m sea-level fall.

Landmark catalogue / 1 of 4

01 / Ferric Sea / impact basin

64.987 latitude, 1.817 longitude. Ancient impact; iron-rich sediment and central uplift

02 / Northern Divide

36.819 latitude, -6.658 longitude. Enormous folded belt; multiple spines and high plateaus

03 / Panterran Plateau

23.000 latitude, -5.000 longitude. Steppe upland above the drowned reservoir valley

04 / Great Reservoir

11.504 latitude, -10.830 longitude. Flooded continental basin, maintained by the dam

05 / Great Panterran Dam

5.133 latitude, -2.476 longitude. Reservoir outlet at a bedrock gorge

06 / Alluvial trade outlet

-4.526 latitude, 10.327 longitude. Raised alluvial terrace beside the downstream eastern port and trade outlet

07 / Western old-growth country

20.000 latitude, -36.000 longitude. Broad Seattle forest country; old growth covers the western watershed and hills

08 / Western Gate

30.249 latitude, -46.788 longitude. Lower mountain saddle; migration checkpoint

09 / Central Gate

35.517 latitude, 22.508 longitude. Second low passage through Northern Divide

10 / Eastern Gate

38.115 latitude, 64.587 longitude. Eastern coastal and ridge crossing

Landmark catalogue / 2 of 4

11 / Southern eastern coast

-37.000 latitude, -86.000 longitude. Wet spring coast beneath the old escarpment

12 / Southern escarpment

-43.000 latitude, -91.000 longitude. Old uplifted margin follows the redesigned eastern coast

13 / Panterran shelf approach

-1.000 latitude, -57.000 longitude. Flooded final approach of the proposed cat migration route

14 / Pieter's fishing village

5.147 latitude, -2.443 longitude. Local dam inset site; terrace location is below global texture resolution

15 / Ancient avian cliff ruins

23.627 latitude, -47.014 longitude. Ruins above the western forest corridor

16 / London / lakeshore country

8.306 latitude, -13.931 longitude. Proposed city on the southwestern lake shore; elevated above the reservoir

17 / Merchant island

-31.190 latitude, 7.219 longitude. Milan shorthand; trade controls in southern strait

18 / Great leeward desert

-5.948 latitude, 115.622 longitude. Dry interior on the left of the marked far-side view

19 / Eastern rift gulf

8.015 latitude, 88.802 longitude. Fault-bounded gulf opening into the southern sea

20 / Sparse eastern archipelago

-24.965 latitude, -68.450 longitude. Two large rifted fragments, one middle island and two small remnants; wide open straits toward Panterra

Landmark catalogue / 3 of 4

21 / Southern shield

-38.000 latitude, -106.000 longitude. Asymmetric dry continental interior

22 / Eastern collision knot

31.303 latitude, 69.314 longitude. Broad transpressional uplands joining eastern terranes

23 / Northern remnant island

55.000 latitude, -161.000 longitude. Small isolated glacial remnant; cooling into winter

24 / Seattle / western forest country

13.500 latitude, -33.400 longitude. Large forested region and lower-river port downstream of the western overflow

25 / Rome / eastern lake shore

10.679 latitude, -2.593 longitude. Proposed high-bank city overlooking the lake; distinct from the downstream metropolis

26 / Deep separating ocean

-10.000 latitude, 182.000 longitude. No island route between eastern and southern continents

27 / Drowned migration shelf

-14.135 latitude, -58.550 longitude. Submerged continental saddles between sparse island peaks

28 / Arctic crown islands

80.000 latitude, -80.000 longitude. Glaciated bedrock islands and straits amid broken pack ice

29 / South polar ice continent

-77.000 latitude, 105.000 longitude. Asymmetric land ice, coastal mountains and outlet glaciers

30 / Southern embayment

-76.000 latitude, -25.000 longitude. Deep marine inlet interrupts the south polar ice sheet

Landmark catalogue / 4 of 4

31 / Northern plateau falls

17.555 latitude, -14.885 longitude. Upper-plateau river drops about 400 m into the reservoir valley

32 / Natural western overflow

12.850 latitude, -20.500 longitude. High bedrock sill; reservoir spills west toward Seattle

33 / Northern interior massifs

48.000 latitude, -8.000 longitude. Broad high country with parallel ranges, glaciated headwalls and deep valleys

34 / Western high valleys

40.000 latitude, -30.000 longitude. Intermontane basins interrupt the vast northern mountain province

35 / Southwestern mountain country

-8.000 latitude, -24.000 longitude. Broad older uplifts, high valleys and cultivated foothills

36 / Southeastern peninsula ranges

-15.000 latitude, 7.000 longitude. Rugged southern spine with lower tributary catchments

Specifications and camera index

Model scale

Radius 6,100 km; axial tilt 35 degrees; orbital period 4,000 Earth years; assumed rotation 24 hours. Globe radius in Blender is 10 units. Physical elevation is separate from 10x display relief. The dam model uses 1 unit = 1 km.

Resolution

1,181,953 globe vertices and 1,179,648 quads. Four packed maps: 8192 x 4096 colour and three 4096 x 2048 physical maps. Thirteen 2048 x 2048 globe renders, two dam renders and one waterfall render at 2400 x 1600. Thirty-six named geographic anchors. Reservoir footprint: 1,417,941 km2. Peak physical elevation: 10.30 km.

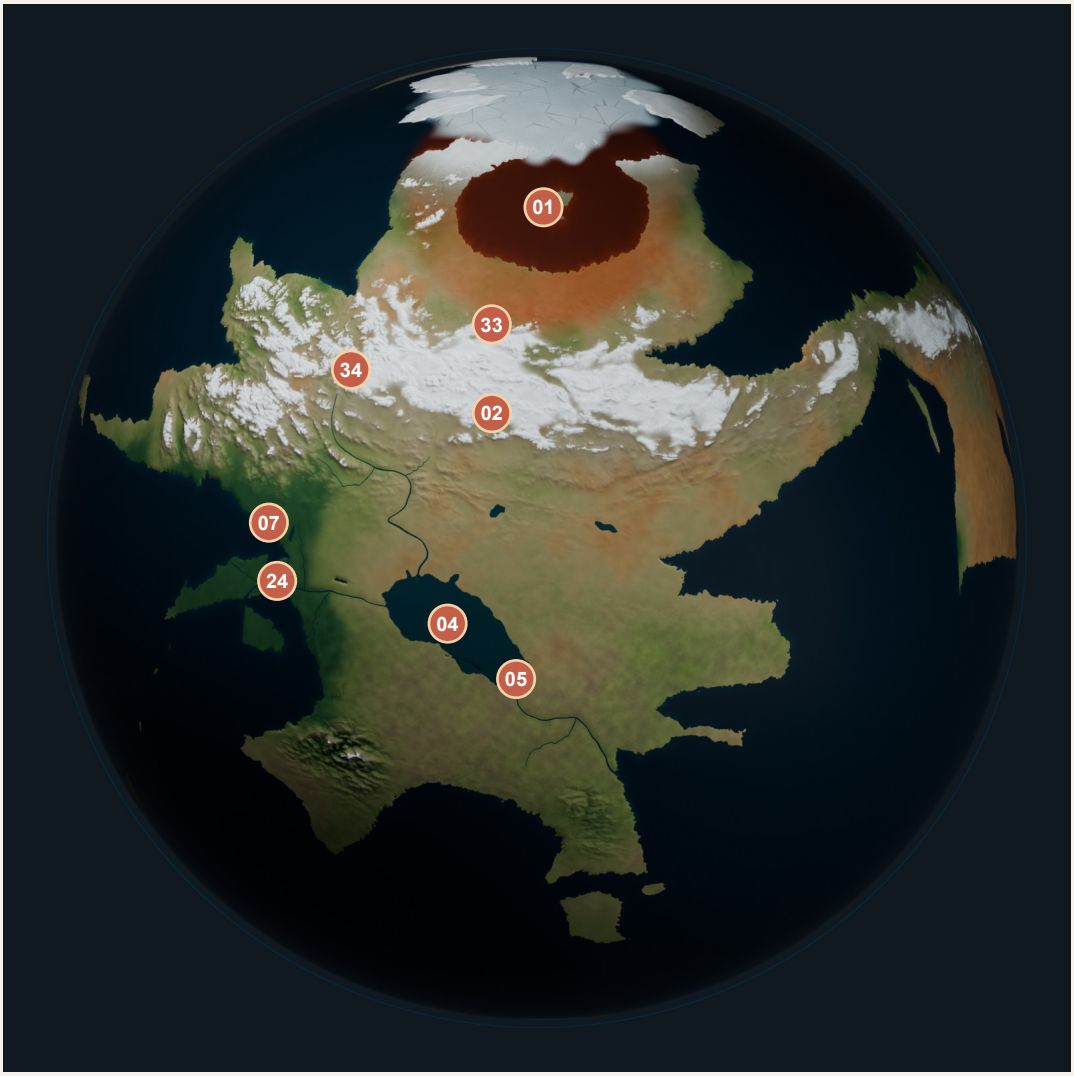
Validation

Panterra and the east are connected; the southern core is separate. The designed shelf connects at -65 m sea level; polar land remains separate. The northern feeder and both reservoir outlets descend. Named city sites are checked against land and reservoir masks. Source scripts, image sizes, gallery links and packed textures are checked. These checks do not replace tectonic, climate, erosion or hydraulic simulation.

Frame 01 - 01_panterra - center 22 lat / 0 lon
Frame 02 - 02_eastern - center 0 lat / 90 lon
Frame 03 - 03_western - center 0 lat / -90 lon
Frame 04 - 04_far_side - center -12 lat / 180 lon
Frame 05 - 05_north_pole - center 90 lat / 0 lon
Frame 06 - 06_south_pole - center -90 lat / 0 lon
Frame 07 - 07_panterra_close - center 15 lat / -8 lon
Frame 08 - 08_impact_close - center 65 lat / 2 lon
Frame 09 - 11_maritime_continent - center -39 lat / -102 lon
Frame 10 - 12_eastern_continent - center -2 lat / 129 lon
Frame 11 - 13_seattle_waterway - center 15 lat / -27 lon
Frame 12 - 14_archipelago - center -20 lat / -65 lon
Frame 13 - 15_plateau_hydrology - center 23 lat / -21 lon

Dam file: frame 1 = valley context; frame 2 = structural detail. Clean PNGs, annotated PNGs, editable models, textures, source scripts, CSV landmarks and this guide are included in the full archive.

Panterra / front hemisphere



Center 22 latitude / 0 longitude. Orthographic view; relief displayed at 10x.

01 / Ferric Sea / impact basin
Ancient impact; iron-rich sediment and central uplift

02 / Northern Divide
Enormous folded belt; multiple spines and high plateaus

04 / Great Reservoir
Flooded continental basin, maintained by the dam

05 / Great Panterran Dam
Reservoir outlet at a bedrock gorge

07 / Western old-growth country
Broad Seattle forest country; old growth covers the western watershed and hills

24 / Seattle / western forest country
Large forested region and lower-river port downstream of the western overflow

33 / Northern interior massifs
Broad high country with parallel ranges, glaciated headwalls and deep valleys

34 / Western high valleys
Intermontane basins interrupt the vast northern mountain province

Eastern hemisphere



Center 0 latitude / 90 longitude. Orthographic view; relief displayed at 10x.

10 / Eastern Gate
Eastern coastal and ridge crossing

19 / Eastern rift gulf
Fault-bounded gulf opening into the southern sea

18 / Great leeward desert
Dry interior on the left of the marked far-side view

22 / Eastern collision knot
Broad transpressional uplands joining eastern terranes

REFERENCE PLATE 03

Western hemisphere



Center 0 latitude / -90 longitude. Orthographic view; relief displayed at 10x.

07 / Western old-growth country
Broad Seattle forest country; old growth covers the western watershed and hills

11 / Southern eastern coast
Wet spring coast beneath the old escarpment

13 / Panterran shelf approach
Flooded final approach of the proposed cat migration route

21 / Southern shield
Asymmetric dry continental interior

27 / Drowned migration shelf
Submerged continental saddles between sparse island peaks

08 / Western Gate
Lower mountain saddle; migration checkpoint

12 / Southern escarpment
Old uplifted margin follows the redesigned eastern coast

20 / Sparse eastern archipelago
Two large rifted fragments, one middle island and two small remnants; wide open straits toward Panterra

24 / Seattle / western forest country
Large forested region and lower-river port downstream of the western overflow

Far-side hemisphere



Center -12 latitude / 180 longitude. Orthographic view; relief displayed at 10x.

12 / Southern escarpment
Old uplifted margin follows the redesigned eastern coast

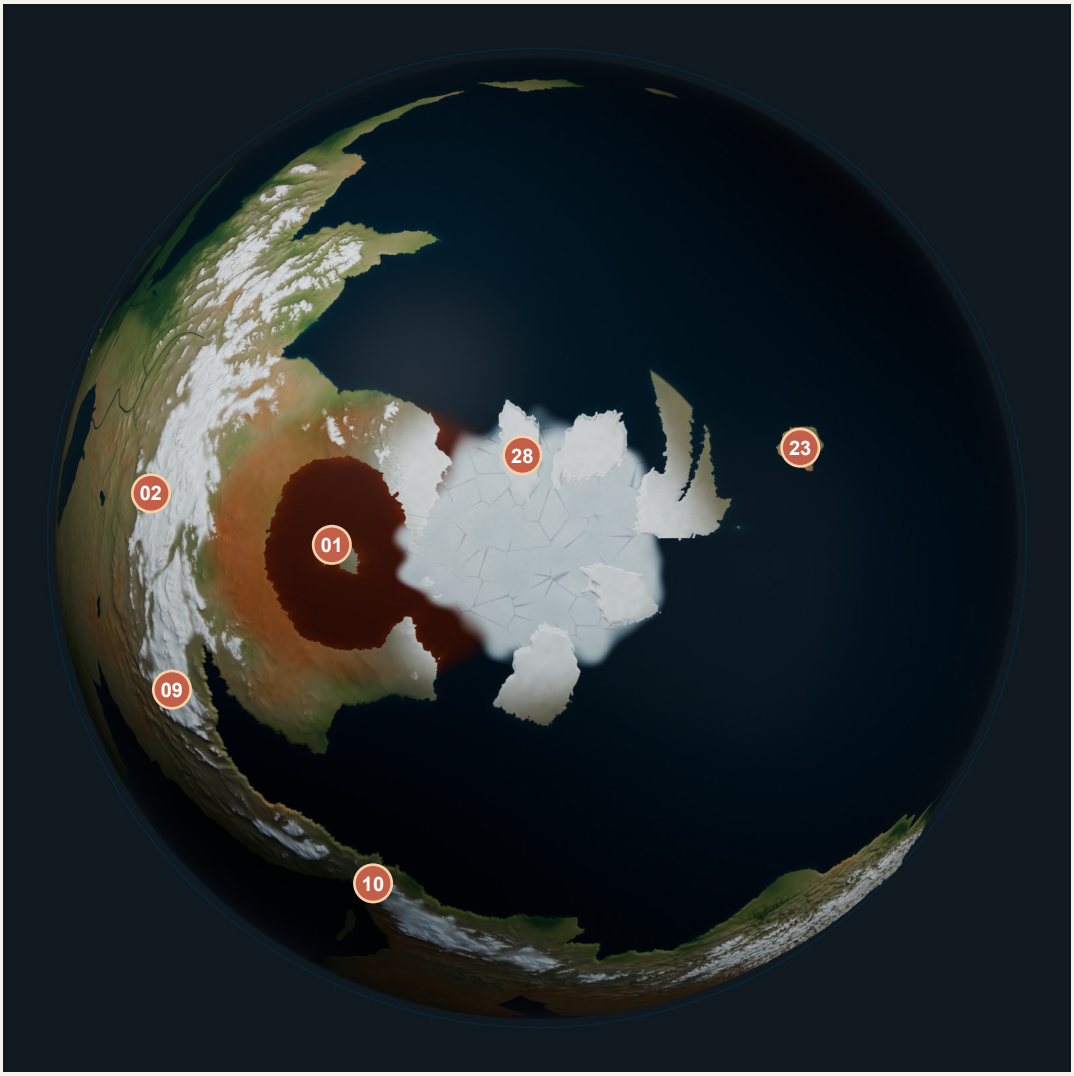
18 / Great leeward desert
Dry interior on the left of the marked far-side view

21 / Southern shield
Asymmetric dry continental interior

23 / Northern remnant island
Small isolated glacial remnant; cooling into winter

26 / Deep separating ocean
No island route between eastern and southern continents

North polar view



Center 90 latitude / 0 longitude. Orthographic view; relief displayed at 10x.

01 / Ferric Sea / impact basin
Ancient impact; iron-rich sediment and central uplift

02 / Northern Divide
Enormous folded belt; multiple spines and high plateaus

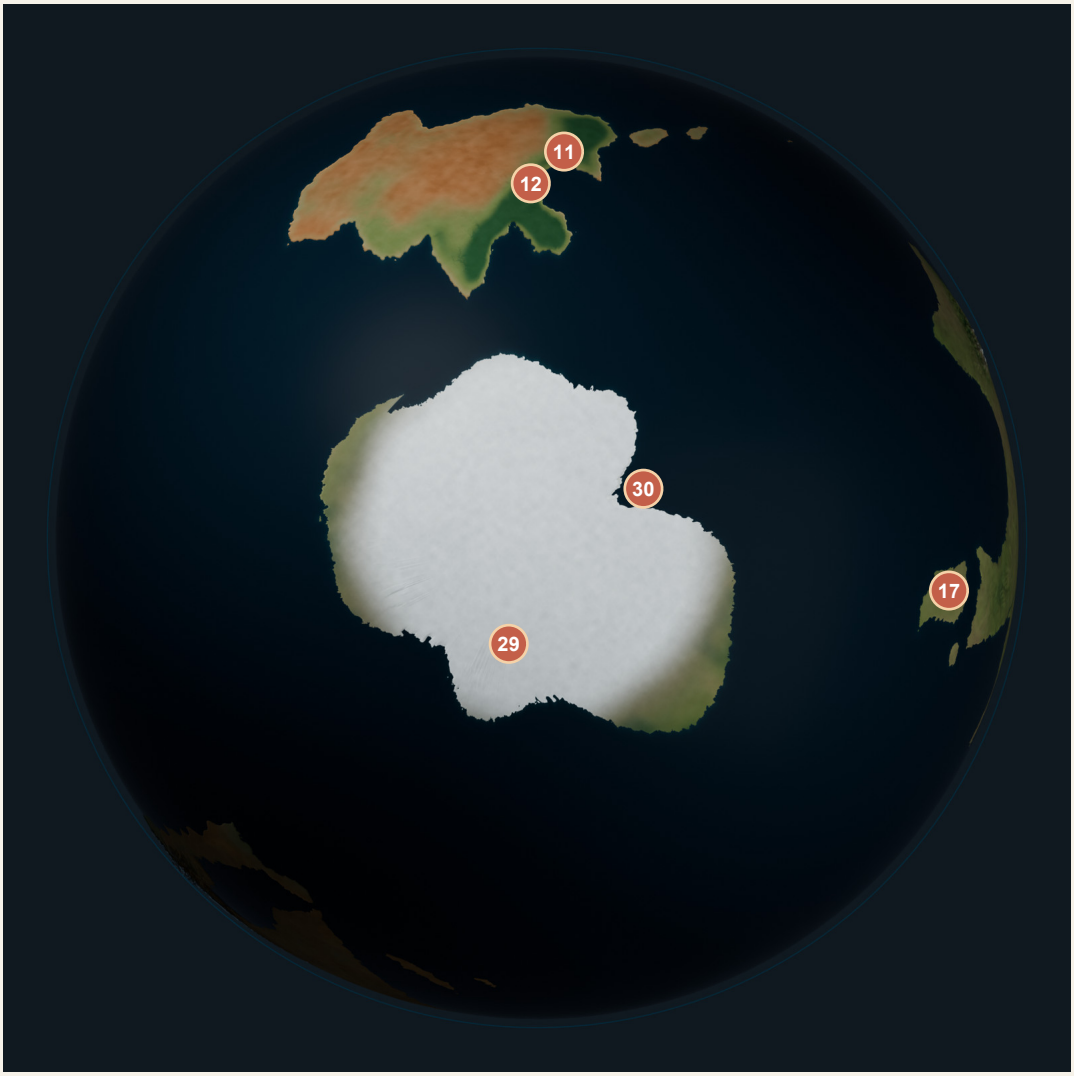
09 / Central Gate
Second low passage through Northern Divide

10 / Eastern Gate
Eastern coastal and ridge crossing

23 / Northern remnant island
Small isolated glacial remnant; cooling into winter

28 / Arctic crown islands
Glaciated bedrock islands and straits amid broken pack ice

South polar view



Center -90 latitude / 0 longitude. Orthographic view; relief displayed at 10x.

11 / Southern eastern coast
Wet spring coast beneath the old escarpment

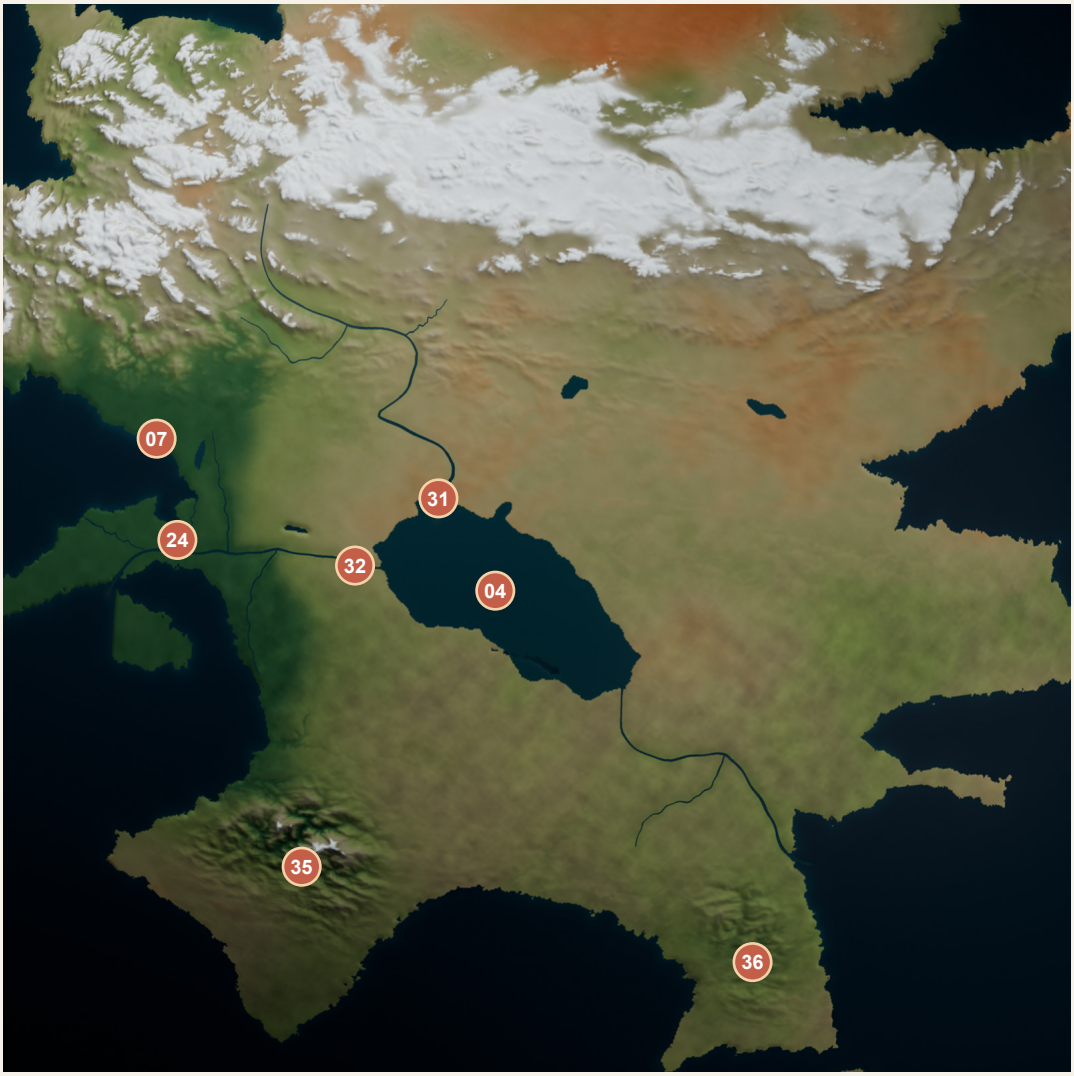
12 / Southern escarpment
Old uplifted margin follows the redesigned eastern coast

17 / Merchant island
Milan shorthand; trade controls in southern strait

29 / South polar ice continent
Asymmetric land ice, coastal mountains and outlet glaciers

30 / Southern embayment
Deep marine inlet interrupts the south polar ice sheet

Panterra / regional study



Center 15 latitude / -8 longitude. Orthographic view; relief displayed at 10x.

FLOW: Northern feeder -> falls -> lake; west -> Seattle; east -> dam

04 / Great Reservoir
Flooded continental basin, maintained by the dam

24 / Seattle / western forest country
Large forested region and lower-river port downstream of the western overflow

32 / Natural western overflow
High bedrock sill; reservoir spills west toward Seattle

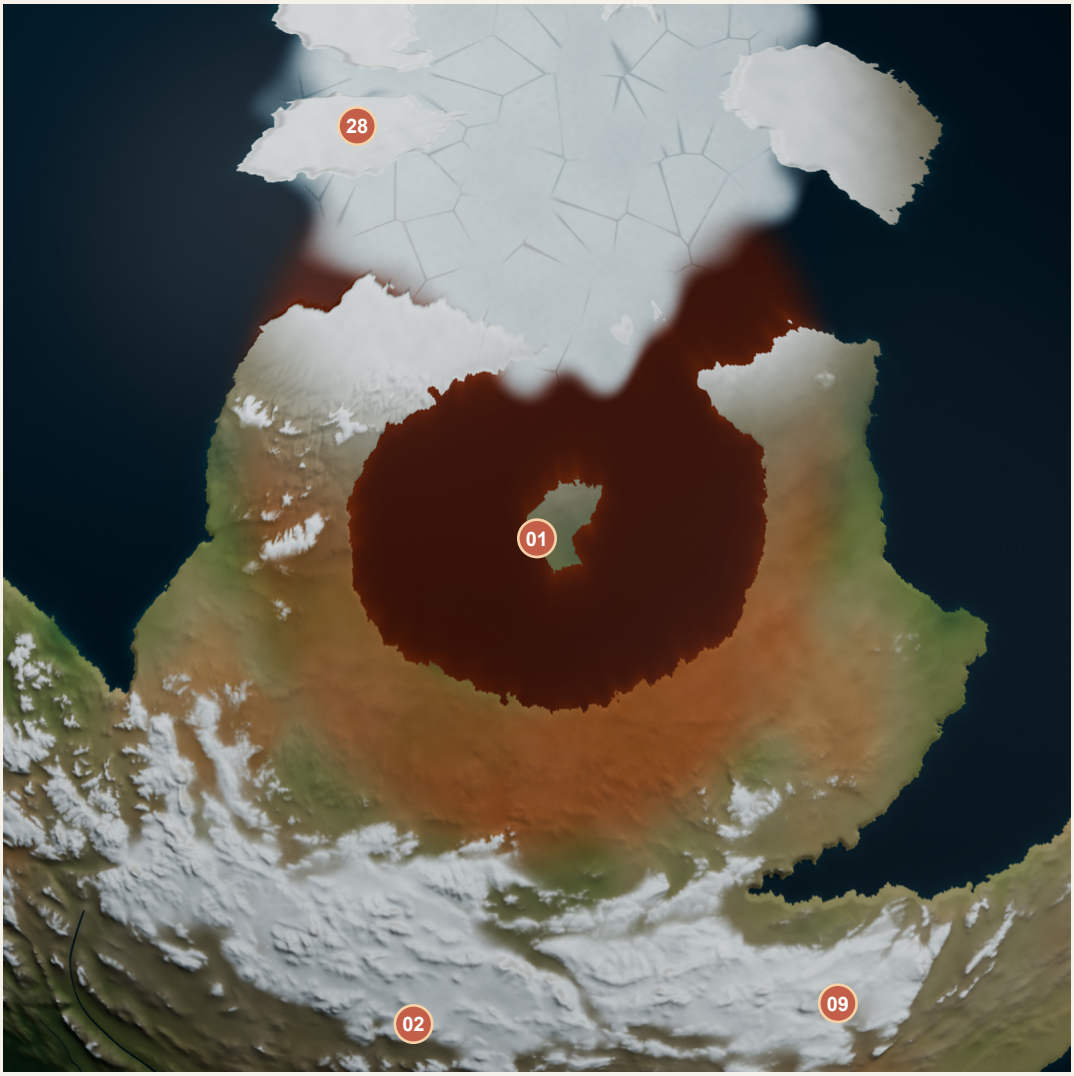
36 / Southeastern peninsula ranges
Rugged southern spine with lower tributary catchments

07 / Western old-growth country
Broad Seattle forest country; old growth covers the western watershed and hills

31 / Northern plateau falls
Upper-plateau river drops about 400 m into the reservoir valley

35 / Southwestern mountain country
Broad older uplifts, high valleys and cultivated foothills

Northern impact basin



Center 65 latitude / 2 longitude. Orthographic view; relief displayed at 10x.

01 / Ferric Sea / impact basin
Ancient impact; iron-rich sediment and central uplift

02 / Northern Divide
Enormous folded belt; multiple spines and high plateaus

09 / Central Gate
Second low passage through Northern Divide

28 / Arctic crown islands
Glaciated bedrock islands and straits amid broken pack ice

Great dam / valley context



Local engineering concept. Scale: 1 unit = 1 km.

A / Great Reservoir

B / Gravity dam / service crest

C / Gated spillway

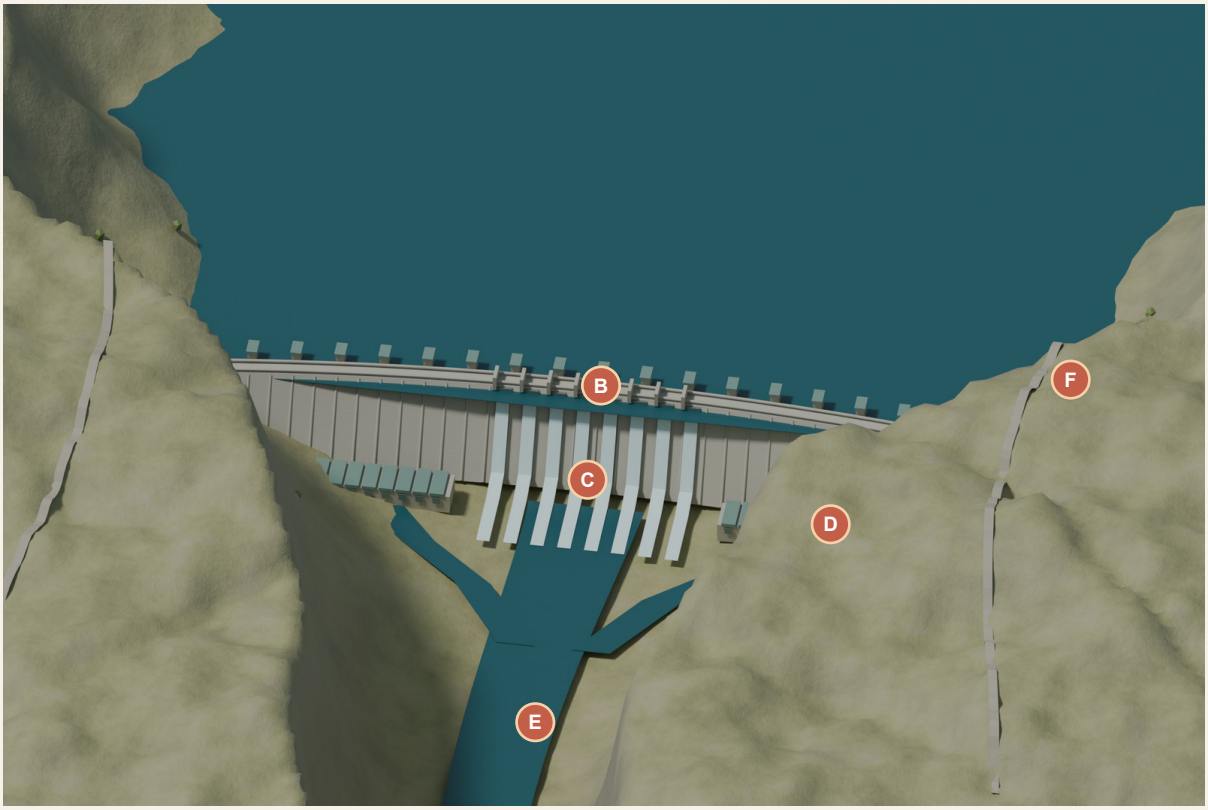
D / Turbine galleries

E / Tailwater / river

F / Abutment checkpoint

G / Fishing village

Great dam / structural study



Local engineering concept. Scale: 1 unit = 1 km.

B / Gravity dam / service crest

C / Gated spillway

D / Turbine galleries

E / Tailwater / river

F / Abutment checkpoint

Southern continent



Center -39 latitude / -102 longitude. Orthographic view; relief displayed at 10x.

11 / Southern eastern coast
Wet spring coast beneath the old escarpment

12 / Southern escarpment
Old uplifted margin follows the redesigned eastern coast

20 / Sparse eastern archipelago
Two large rifted fragments, one middle island and two small remnants; wide open straits toward Panterra

21 / Southern shield
Asymmetric dry continental interior

27 / Drowned migration shelf
Submerged continental saddles between sparse island peaks

Eastern continent



Center -2 latitude / 129 longitude. Orthographic view; relief displayed at 10x.

10 / Eastern Gate
Eastern coastal and ridge crossing

19 / Eastern rift gulf
Fault-bounded gulf opening into the southern sea

18 / Great leeward desert
Dry interior on the left of the marked far-side view

22 / Eastern collision knot
Broad transpressional uplands joining eastern terranes

Seattle / primary waterway



Center 15 latitude / -27 longitude. Orthographic view; relief displayed at 10x.

FLOW: Northern feeder -> falls -> lake; west -> Seattle; east -> dam

04 / Great Reservoir

Flooded continental basin, maintained by the dam

24 / Seattle / western forest country

Large forested region and lower-river port downstream of the western overflow

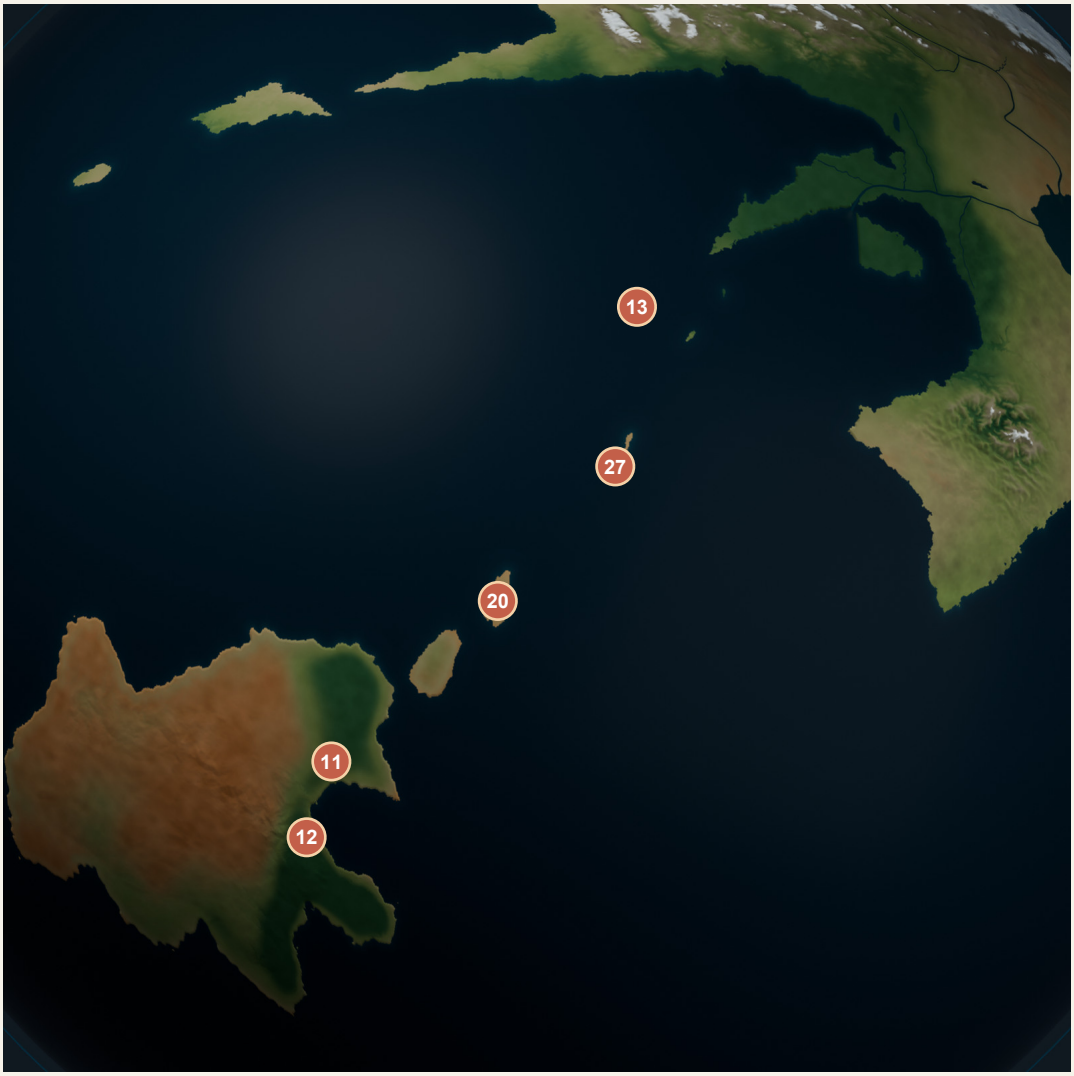
07 / Western old-growth country

Broad Seattle forest country; old growth covers the western watershed and hills

32 / Natural western overflow

High bedrock sill; reservoir spills west toward Seattle

Eastern archipelago



Center -20 latitude / -65 longitude. Orthographic view; relief displayed at 10x.

11 / Southern eastern coast
Wet spring coast beneath the old escarpment

12 / Southern escarpment
Old uplifted margin follows the redesigned eastern coast

13 / Panterran shelf approach
Flooded final approach of the proposed cat migration route

20 / Sparse eastern archipelago
Two large rifted fragments, one middle island and two small remnants; wide open straits toward Panterra

27 / Drowned migration shelf
Submerged continental saddles between sparse island peaks

PLATEAU / RIVER SYSTEM



Center 23 latitude / -21 longitude. Orthographic view; relief displayed at 10x.

FLOW: Northern feeder -> falls -> lake; west -> Seattle; east -> dam

02 / Northern Divide
Enormous folded belt; multiple spines and high plateaus

03 / Panterran Plateau
Steppe upland above the drowned reservoir valley

04 / Great Reservoir
Flooded continental basin, maintained by the dam

05 / Great Panterran Dam
Reservoir outlet at a bedrock gorge

24 / Seattle / western forest country
Large forested region and lower-river port downstream of the western overflow

31 / Northern plateau falls
Upper-plateau river drops about 400 m into the reservoir valley

32 / Natural western overflow
High bedrock sill; reservoir spills west toward Seattle

NORTHERN PLATEAU FALLS



Local landform blockout. True vertical scale; 1 unit = 1 km. Water is not fluid-simulated.

A / Upper plateau river / 1090 m
Local study; true vertical scale

B / Main waterfall / 250 m
Local study; true vertical scale

C / Plunge pool / 840 m
Local study; true vertical scale

D / Lower cascades
Local study; true vertical scale

E / Reservoir valley / 690 m
Local study; true vertical scale

F / Steppe plateau
Local study; true vertical scale